Generating Labeled Data: Evaluating the Performance of cGANs and Diffusion Models for Image Synthesis

Master's Thesis in Computational Engineering

submitted
by

Shubhi Sharma

born May 9, 1995 in India

Written at

Lehrstuhl für Mustererkennung (Informatik 5)
Department Informatik
Friedrich-Alexander-Universität Erlangen-Nürnberg.

Advisor: Hakan Calim

Started: Oct. 21, 2024

Finished: April 22, 2025

Erklärung

Ich versichere, dass ich die Arbeit ohne fremde Hilfe und ohne Benutzung anderer als der angegebenen Quellen angefertigt habe und dass die Arbeit in gleicher oder ähnlicher Form noch keiner anderen Prüfungsbehörde vorgelegen hat und von dieser als Teil einer Prüfungsleistung angenommen wurde. Alle Ausführungen, die wörtlich oder sinngemäß übernommen wurden, sind als solche gekennzeichnet.

Ort, Datum

Unterschrift

Abstract

Generative AI models have gained prominence in data synthesis in various domains. These models learn the distribution of data and generate new samples according to the training data. GANs, VAEs, transformers, and diffusion models were heavily employed since 2014 because they can produce high-quality images, with open data sets, expensive hardware, and heavy use such as creative image generation, simulation, and data augmentation.

Despite the progress in unsupervised learning, most machine learning issues are still very much dependent on annotated datasets for structure and interpretability. This is particularly critical in assistive technology, such as visually impaired navigation systems, where accurate scene understanding is critical. Common datasets such as Cityscapes, however, don't provide images with centered sidewalks—a basic necessity for front-facing navigation. This thesis addresses the limitation by utilizing generative models to enhance the SENSATION dataset, which is focused on semantically annotated pedestrian scenes with centrally aligned sidewalks. The underrepresented classes are also balanced with a two-stage data enhancement pipeline before using generative modeling.

SENSATION's segmentation masks were used to guide image generation, with semantic structure preserved and visual content diversity introduced. Two of the newest models—Pix2PixHD (a cGAN) and Stable Diffusion with ControlNet—were explored for this. Different architectural and training methods were experimented with, such as fine-tuning, layer freezing, and parameter adjustment.

While Pix2PixHD struggled to handle sparse training data and transfer learning constraints, the Stable Diffusion + ControlNet pipeline consistently produced semantically accurate and photo-realistic images. Quantitative and qualitative evaluations ensured that models based on diffusion yield a stronger solution to enhance datasets towards assistive navigation.

Kurzfassung

Generative KI-Modelle haben bei der Datensynthese in verschiedenen Bereichen an Bedeutung gewonnen. Diese Modelle lernen die Verteilung von Daten und generieren anhand der Trainingsdaten neue Muster. GANs, VAEs, Transformatoren und Diffusionsmodelle werden seit 2014 in großem Umfang eingesetzt, da sie qualitativ hochwertige Bilder erzeugen können, und zwar mit offenen Datensätzen, teurer Hardware und intensiver Nutzung wie kreativer Bilderzeugung, Simulation und Datenerweiterung.

Trotz der Fortschritte im Bereich des unüberwachten Lernens sind die meisten Probleme des maschinellen Lernens immer noch sehr stark von annotierten Datensätzen abhängig, um Struktur und Interpretierbarkeit zu gewährleisten. Dies ist besonders wichtig für Hilfstechnologien, wie z. B. Navigationssysteme für Sehbehinderte, bei denen ein genaues Verständnis der Szene entscheidend ist. Gängige Datensätze wie Cityscapes liefern jedoch keine Bilder mit zentrierten Bürgersteigen - eine Grundvoraussetzung für die frontale Navigation. In dieser Arbeit wird diese Einschränkung durch den Einsatz generativer Modelle zur Verbesserung des SENSATION-Datensatzes behoben, der sich auf semantisch annotierte Fußgängerszenen mit mittig ausgerichteten Gehwegen konzentriert. Die unterrepräsentierten Klassen werden ebenfalls mit einer zweistufigen Datenanreicherungspipeline ausgeglichen, bevor die generative Modellierung eingesetzt wird.

Die Segmentierungsmasken von SENSATION wurden zur Steuerung der Bilderzeugung verwendet, wobei die semantische Struktur erhalten blieb und die Vielfalt der visuellen Inhalte eingeführt wurde. Zwei der neuesten Modelle - Pix2PixHD (ein cGAN) und Stable Diffusion mit ControlNet - wurden hierfür untersucht. Es wurde mit verschiedenen Architektur- und Trainingsmethoden experimentiert, wie z. B. Feinabstimmung, Einfrieren von Schichten und Parameteranpassung.

Während Pix2PixHD Schwierigkeiten hatte, mit spärlichen Trainingsdaten und Einschränkungen beim Transferlernen umzugehen, produzierte die Stable Diffusion + ControlNet-Pipeline durchweg semantisch genaue und fotorealistische Bilder. Quantitative und qual-

itative Auswertungen haben gezeigt, dass auf Diffusion basierende Modelle eine bessere Lösung zur Verbesserung von Datensätzen für die assistive Navigation bieten.

Acknowledgments

I am deeply grateful to Prof. Dr.-Ing. habil. Andreas Maier for providing me with an invaluable opportunity to pursue this research. His guidance and insights have been pivotal in my studies and have profoundly shaped my academic journey.

I am immensely grateful to Mr. Hakan Calim for their constant support and expert advice throughout my thesis. Their expertise and encouragement have been crucial in helping me navigate research challenges. A special thanks to the HPC cluster support team for assistance throughout the experimental phase.

I could not have completed this thesis without these exceptional people's help, advice, and encouragement. I am grateful for their time, patience, and belief in me.

Contents

Chapter 1

Motivation and Introduction

1.1 Motivation

Real-world machine learning applications, especially those that involve safety-critical decision-making, often rely on large, well-annotated datasets to function effectively. Assistive technology is a good case in point, where vision-based navigation systems assist Visually Challenged Individuals (VCIs) to move on pedestrian sidewalks. For such systems, an accurate understanding of pedestrian paths, particularly sidewalks, is essential. Detection of obstacles, traffic signs, traffic lights, uneven paths is of immense importance.

But public datasets such as Cityscapes despite capturing road scenes are not sidewalk-focused but are road traffic focused instead. This matters in order to mirror a forward-looking view that would be natural for a VCI.[1] This limits the effectiveness of training of perception systems for deployment in the real world.

At the same time, collecting and annotating new datasets for such specialized tasks is expensive and time-consuming. Traditional data augmentation techniques too come with their limitation to the diversity in new data that they can produce.[2] This motivates the exploration of synthetic dataset expansion through the assistance of generative models capable of completing existing annotated data without losing semantic structure.[3] As a precursor step to training navigation guidance systems which can provide audio and/or haptic form of guidance, these synthesized labeled datasets can prove to be of great advantage.[3]

1.2 Introduction

Generative AI models have proven to be powerful tools for realistic data synthesis across different modalities. By training a conditional generative adversarial network (GAN) in a related but data-rich source domain, it is possible to learn a model that captures a broader range of invariances. This learned model can then be transferred to a low-data target domain, where the synthesized data helps facilitate effective neural network training despite limited original samples.[3] In computer vision, they have been used for image editing and generation, simulation, and data augmentation.[4] At the same time, transfer learning in its many variations has also proven to be useful in data-scarce cases.

Since 2014, Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), diffusion models, and transformers have taken center stage due to their ability to produce high-quality data, thanks to hardware developments, open data sets, and scalable training procedures. Diffusion models have over the last few years emerged as players to be reckoned with in producing semantically rich and photorealistic outputs with good diversity.

This thesis explores the feasibility of using generative models like Pix2PixHD[5], SPADE[6], ControlNet[7], DDPM[8] to synthetically expand a labeled dataset with an emphasis on blind visual navigation. Relying on the application of segmentation maps as structure guides, here one aims to synthesize novel images maintaining spatial semantics of passageways without modifying their look. In order to improve results, transfer learning has also been experimented with. A new aspect introduced and tested in this thesis is the usage of poorly formed outputs from fine tuning pix2pixhd model as inputs to controlnet diffusion model. My hypothesis behind this experimentation used in [7], that is, the kind of input masks that they used for image generation weren't simple boundaries but some textural level guidance too provided in the mask. This implied that the more the guidance in the mask, the more would be the detailing in the generated image. This led to the originally planned comparison of cGANs and diffusion techniques into comparison and integration of theirs. The integration resulted in improving the individual performances of the models in isolation.

The experiments expanded the initial dataset significantly and balanced the under-represented classes with photorealistic images. The experiments also showed how with increasing textural patterns in the training image/mask, the ability of ControlNet diffusion model to produce better images increased. This can be used to improve the training of navigation guidance

1.3 Problem Formulation

The primary challenge addressed in this thesis is the scarcity and imbalance of annotated training data for vision-based navigation systems created for VCIs. Specifically, the SENSATION dataset, while focused on sidewalk-centered scenes, suffers from under-representation of key semantic classes such as crosswalks, obstacles, and signage. Additionally, the segmentation masks associated with the dataset are fixed, limiting the diversity of visual patterns that can be modeled.

To address this, this study explores a generative approach to dataset augmentation. Given a semantic segmentation mask M from the SENSATION dataset, the goal is to synthesize multiple photorealistic image variants $\{I_1, I_2, \ldots, I_n\}$ that preserve the semantic layout but introduce sufficient visual diversity. These new samples should enhance the dataset's class balance and improve generalization in downstream models.

Formally, we define the task as learning a conditional generative function $G : (M, z) \rightarrow I$, where M is the input mask, z is a noise or condition vector, and I is the generated image. This function must:

- Maintain structural fidelity to the input mask.
- Introduce diversity in texture, lighting, and visual context.
- Produce outputs that are semantically valid and realistic enough for training segmentation or detection models.

To this end, two generative paradigms were investigated: a cGAN(Pix2PixHD) and diffusion-based models Stable Diffusion + ControlNet, DDPM and compared based on diversity and quality of images produced. These were trained and/or fine-tuned on the SENSATION dataset with the aim of expanding the dataset's diversity and addressing underrepresented semantic categories. Fine-tuning was carried out on models that were previously trained on Cityscapes for Pix2PixHD, ADE20k for DDPM as well as stable diffusion.

1.4 Outline

This thesis is structured into several chapters dedicated to a specific research segment. The research focuses on synthesizing images from their semantic segmentation masks using cGAN and diffusion model and comparing them to achieve this goal. Each chapter provides an in-depth exploration of a particular aspect of the research, offering detailed analysis and findings. Below is a comprehensive overview of each chapter, providing a detailed summary of the content covered in each section. :

Chapter 2: Literature Review

This chapter provides an overview of the existing research and providing fundamental knowledge of the key domains used in this thesis and the challenges these methods face. It includes a critical evaluation of generative models particularly cGANs and diffusion models, setting the stage for the research question addressed in this thesis. A major interweaving is the concept of transfer learning.

Chapter 3: Generation of Images Using Semantic Segmentation Masks

The third chapter introduces the concept and methodology of synthetic data generation using various cGAN and diffusion models and how have been leveraged through various methods to expand datasets. This section explains the theoretical foundation for using pix2pixHD and controlnet as the main cGAN and diffusion models used for comparison of diversity and photorealism of the dataset that they create. The chapter also brings in the notion of fine-tuning models trained on similar dataset for further strengthening the models.

Chapter 4: Methodological Foundations: Dataset, Architecture, and Preprocessing

This chapter details the experimental framework designed to evaluate the comparative performance of the aforementioned models. It describes the dataset's selection and preparation, the neural network's specific architecture, the training process, and the criteria for assessing the model's performance.

Chapter 5: Experimentation

This chapter will evaluate the model across different scenarios to assess its performance. The evaluation aims to determine the model's generalizability and effectiveness in accurately predicting decay parameters in diverse settings. This phase is crucial to understanding the model's practical applicability before going into the detailed results and comparisons with traditional methods.

Chapter 6: Results Evaluation And Analysis

This chapter presents the outcomes of the experiments, analyzing the model's capability to generate realistic images out of their semantic masks. The results from the 2 suggested models are compared.

Chapter 7: Conclusion

The concluding chapter synthesizes the study's findings, discussing their implications for expansion of labeled data in data-scarce fields. It also suggests how generative models, especially diffusion models can be guided to generate images of certain desired-structures with control in more than one form, that is, text plus semantic masks. It evaluates the thesis's contribution to the broader field, considers the limitations of the current research, and suggests avenues for further investigation. This chapter also discusses the significant scope for additional research, with each proposed direction poised to enhance the initial work presented in this study.

As presented in 1.4, this structure aims to provide a comprehensive and coherent narrative that gradually unfolds the research process from foundational concepts to practical applications and future directions.

Chapter 2: Literature Review

↓

Chapter 3: Generation of Images Using Semantic Segmentation Masks

↓

Chapter 4: Methodological Foundations: Dataset, Architecture, and Preprocessing

↓

Chapter 5: Experimentation

↓

Chapter 6: Results Evaluation And Analysis

↓

Chapter 7: Conclusion

Chapter 2

Literature Review

This chapter dives into the concepts and fundamentals of the thesis. It starts with the understanding of deep neural networks(DNNs), and proceeds to finally lay forward useful details of the main highlights of the work,i.e., cGANs and diffusion models.

2.1 Deep Neural Networks (DNNs)

Conventional machine learning models utilize manually-crafted features tailored to a specific task, whereas deep learning systems are capable of extracting useful features automatically from raw inputs[9] as can be observed in 2.1. A typical Neural Network (NN) is composed of numerous interconnected processing units, or neurons, which generate real-valued outputs. Earlier NN-based models, often limited to only a few layers, have existed for several decades. A Deep Neural Network (DNN) is an artificial neural network that contains multiple layers between the input and output layers. Complex non-linear relationships can be learned using these layers.[10] DNNs can be trained to discover hierarchical representations in which later layers obtain increasingly abstracted features, allowing the network to identify intricate data structures.[12]The initial training of deep networks suffered from issues such as the problem of the vanishing gradient, which hindered the performance of deeper architecture until the appearance of solutions including improved activation functions and initialization methods. This resurgence was mainly fueled by architectural advancements such as the use of ReLU activation functions and techniques like dropout, which enabled efficient training of deeper networks.[2]

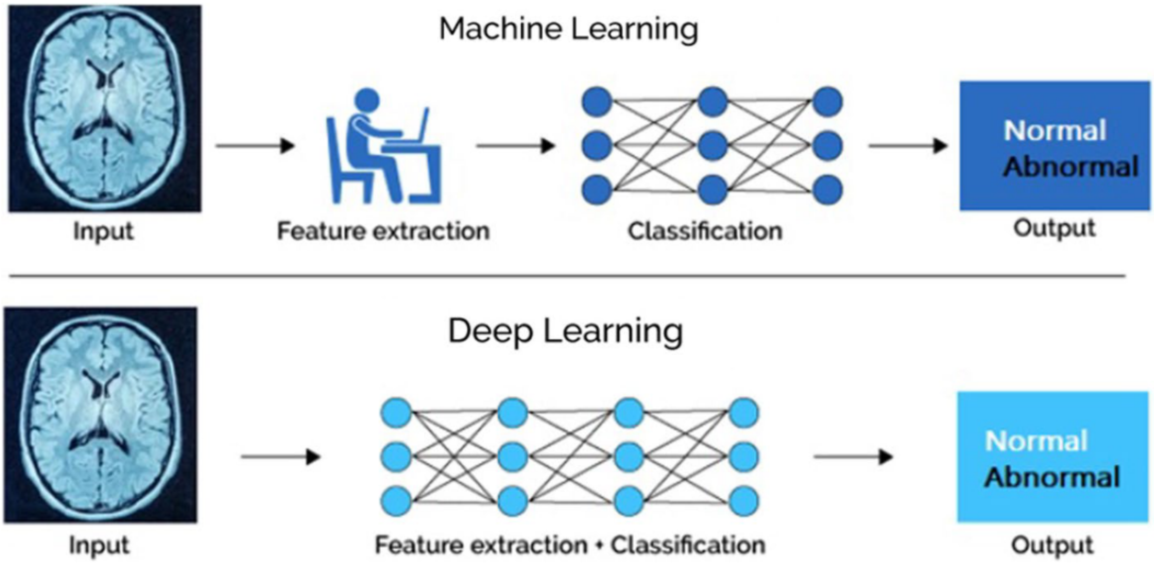


Figure 2.1: Tradition Machine Learning versus Deep Learning [11]

2.2 Labeled and Unlabeled Data

2.2.1 Difference Between Supervised and Unsupervised Learning Methods

Based on the type of data available during training, machine learning techniques can be broadly classified into supervised and unsupervised learning. Other important paradigms include self-supervised, semi-supervised and reinforcement learning. In supervised learning, the goal is to learn a mapping from inputs $\mathbf{x}$ to outputs $\mathbf{t}$ from a labeled set of input-output pairs. This approach is mostly used for Machine Learning (ML) tasks like classification, regression and semantic segmentation. In unsupervised learning, we have an input corresponding to $\mathbf{x}$ but no target values and the goal is to discover interesting structure in the data.This technique is used for uncovering hidden patterns, structures and/or relationships within the data like in clustering, dimensionality reduction, and density estimation.[13]

2.2.2 Labeled and Unlabeled Data Uses

When input training data $\mathbf{x}$ comes with a target value t, the pair $(\mathbf{x}, t)$ is referred to as labeled data. For example, in a classification task of images, $\mathbf{x}$ would be an image and t would be a class label, say, "apple." Unlabeled data comes with no such label or annotation.

2.3 Data Augmentation

ML and especially deep learning techniques are data hungry and dearth of good quality, large training datasets is a common issue. One popular choice to enlarge and diversify the available datasets is data augmentation. This involves applying systematic transformations to the existing samples of data. In image-based tasks, augmentation is utilized to improve the generalization of a model by making it robust to variations that could exist in real-world data. Commonly applied manual augmentation techniques include geometric transformations such as rotation, flipping, cropping, translation, and scaling.[14] These transformations simulate different spatial arrangements of the same object. Additionally, photometric transformations such as changing brightness, contrast, color saturation, and introducing noise are utilized to mimic lighting or sensor variations. These augmentation techniques don't alter the semantic content of the image but introduce useful variability, allowing the model to learn invariance to some view or distortion without needing additional manual annotations. Fig.2.2 shows some of the above mentioned augmentation methods.

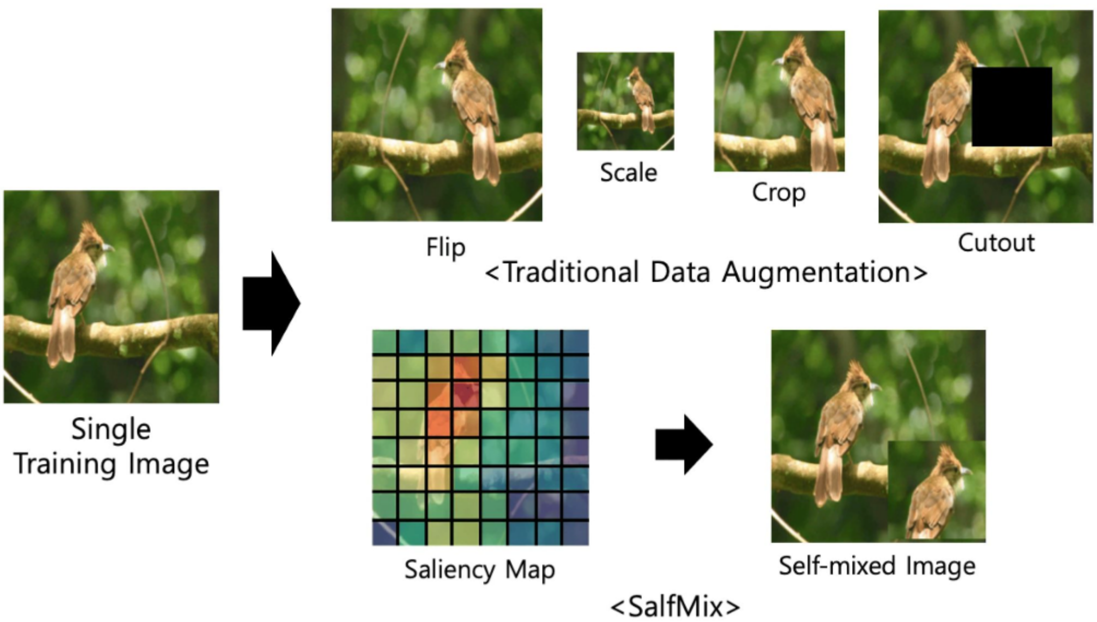


Figure 2.2: Traditional Data Augmentation Techniques
[15]

2.4 AI-based Visual Assistance Systems

A comprehensive review of state-of-the-art assistive systems for the visually-challenged individuals showcases the growing reliance on deep learning to carry out object detection, obstacle avoidance, and semantic scene understanding tasks by quoting notable challenges with respect to dataset diversity and real-time applicability.[16] AI-based visual aid systems, exemplified in Fig.2.3, are designed to assist visually impaired individuals by interpreting the surroundings and presenting useful information in the form of audio or tactile feedback. Such systems tend to employ computer vision, natural language processing, and other machine learning techniques to detect objects, identify scenes, and assist navigation. Compared to traditional Electronic Travel Aid (ETA), which use sonar or infrared sensors to detect nearby obstacles, AI-powered solutions provide more contextual information through analysis of camera feed. With the addition of deep learning models, modern systems can detect objects in real-time, read text, perform semantic segmentation, and even provide route guidance.[17]

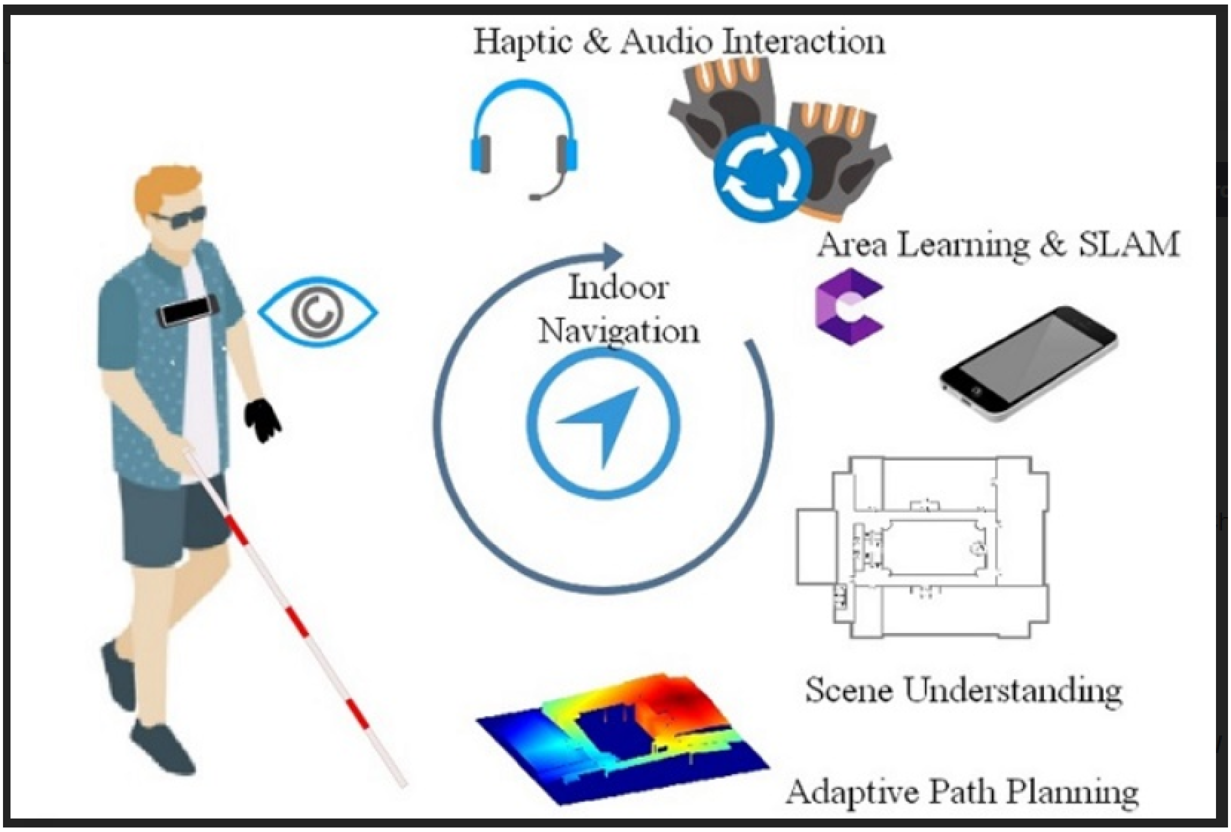


Figure 2.3: Working Mechanism of an Assistive System for the VCIs [18]

2.5 Generative Models

Even though classical graphics algorithms are extremely effective at constructing and manipulating virtual worlds, the process is still time-consuming and costly since it entails the explicit representation of all the material in a scene. If we, however, were to generate photorealistic images from models learned through data, the problem of rendering would then be framed as one of model learning and prediction. This would highly facilitate the construction of virtual worlds by enabling models to learn about scene composition from datasets rather than having it programmed by hand.[5] Generative models such as GANs, VAEs, and Diffusion Models are prominent tools for producing realistic images and learning data distributions. The models can generate novel data samples which resemble samples of a training set and therefore are suitable for usage in applications such as image synthesis, data enhancement, and simulation.

A generative model, in general, attempts to learn the underlying probability distribution $p(x)$ of the data and generate new samples that are statistically similar to the provided data.[19]

2.6 Synthetic Data Generation using Generative Models

While traditional data augmentation techniques, such as photometric and geometric transformations, introduce beneficial variability, they are inherently limited in their ability to generate truly novel data instances. These methods apply fixed, predefined operations and do not add fundamentally new content to the dataset. Recent advancements in generative modeling have enabled the creation of entirely synthetic data, significantly enhancing the diversity and richness of training datasets.[20].

Generative models, including GANs, VAEs, and diffusion models, can learn the underlying data distribution and generate new data points that are not mere augmented copies but novel and semantically meaningful instances. This capability makes generative models valuable tools for synthetic data generation, especially in domains where labeled data is scarce, expensive, or difficult to obtain.[3].

In computer vision, synthetic images produced by trained generative models can serve as additional training data for tasks such as object detection, segmentation, or image classification. Incorporating high-quality synthetic samples into datasets enables models to generalize better and achieve improved performance on real-world data.[20].

Furthermore, these generative approaches can be conditioned on specific inputs—such as segmentation maps, class labels, or textual descriptions—to produce outputs that are explicitly targeted and semantically guided. This makes them particularly suitable for structured data augmentation in tasks requiring pixel-level coherence, such as semantic segmentation.[3].

2.7 Deterministic and Probabilistic Models

2.7.1 Deterministic Model

In deterministic models, the output is entirely determined by the input; for a particular input, the model will always give the same output with no variability.[13]

In mathematical terms, a *deterministic model* defines a fixed function $f : \mathcal{X} \to \mathcal{Y}$, such that for any input $\mathbf{x} \in \mathcal{X}$, the output is always the same[13]:

$$\mathbf{y} = f(\mathbf{x})$$

There is no scope of uncertainty in the output,i.e, the mapping from input to output is exact and repeatable.[13]

2.7.2 Probabilistic Model

A probabilistic model represents uncertainty by putting a distribution over possible outputs, allowing the model to produce varied outputs for the same input depending on to the underlying probability distribution.[13]

In mathematical terms, a *probabilistic model* describes a distribution over possible outputs given an input. It defines either the joint distribution $p(\mathbf{x}, \mathbf{y})$, or more commonly, the conditional distribution:

$$p(\mathbf{y} \mid \mathbf{x})$$

This formulation allows the model to capture uncertainty by assigning probabilities to different outcomes for a given input. For generative tasks, probabilistic models often define a distribution over the data itself, i.e.,

$$\mathbf{x} \sim p_\theta(\mathbf{x}),$$

where θ are learnable parameters of the model.[13], [21]

2.8 Generative Adversarial Nets(GANs) and Conditional GANs(cGANs)

2.8.1 Generative Adversarial Net(GAN)

Introduced by Goodfellow et al.[22] in 2014, GANs represent a class of generative models that formulate the data generation process as a two-player minimax game between a generator and a discriminator. A GAN consists of two neural networks trained simultaneously: a generator G and a discriminator D. The generator maps a random noise vector $\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})$ to a synthetic data instance $G(\mathbf{z})$. The discriminator takes an input $\mathbf{x}$ and outputs a probability $D(\mathbf{x}) \in [0, 1]$, indicating whether the input is real (from the dataset) or fake (from the generator).

The networks are trained using the following minimax objective:

$$\min_G \max_D \mathbb{E}_{\mathbf{x}\sim p_{\text{data}}}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}}}[\log(1 - D(G(\mathbf{z})))].$$

The generator tries to minimize this objective by producing data that the discriminator cannot distinguish from real samples, while the discriminator aims to maximize it by improving its classification accuracy. Fig.2.4 portrays how this minmax game is played within a GAN. Although GANs are capable of producing high-quality outputs, they often suffer from training instability and mode collapse, where the generator produces limited diversity [24].

2.8.2 Conditional Generative Adversarial Net(cGAN)

In a cGAN, the generator learns a conditional mapping $G(\mathbf{z}, \mathbf{c})$ to synthesize outputs based on both random noise and conditioning input, while the discriminator evaluates whether

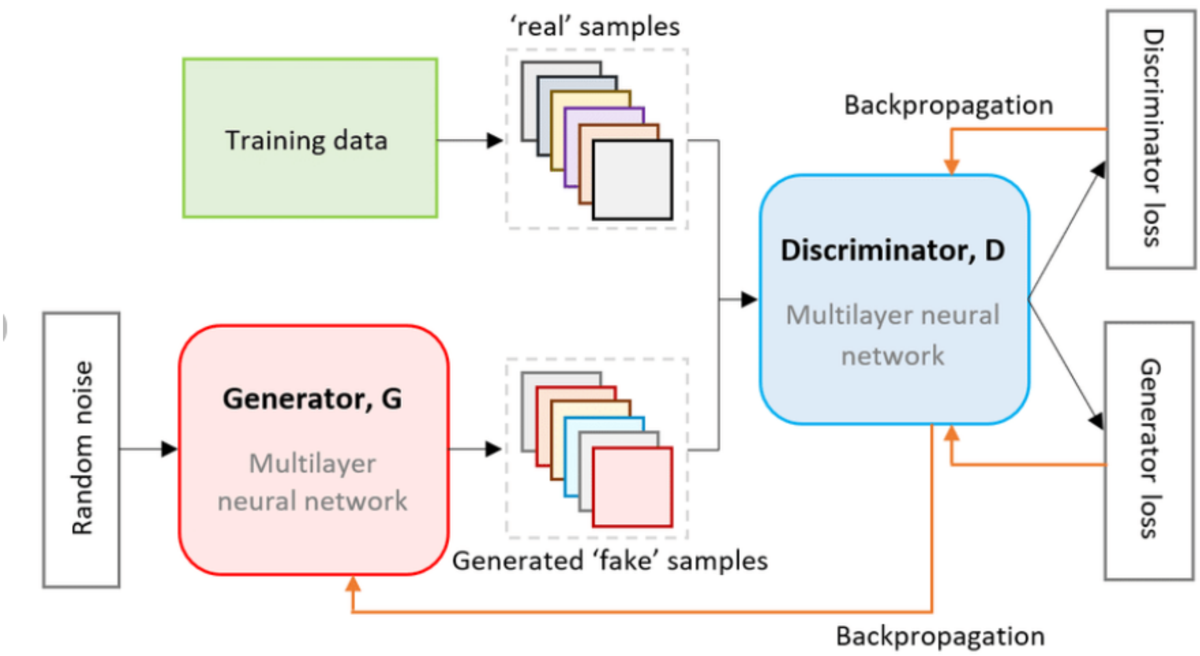


Figure 2.4: GAN Architecture
[23]

the given pair $(\mathbf{x}, \mathbf{c})$ is real or fake. The objective thus becomes:

$$\min_G \max_D \mathbb{E}_{\mathbf{x},\mathbf{c}\sim p_{\text{data}}}[\log D(\mathbf{x}, \mathbf{c})] + \mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}},\mathbf{c}\sim p_{\text{data}}}[\log(1 - D(G(\mathbf{z}, \mathbf{c}), \mathbf{c}))].$$

[4] This conditioning mechanism makes cGANs especially suitable for structured image generation tasks, such as image-to-image translation, semantic segmentation, and super-resolution, where desired outputs depend on specific input configurations[4]. fig.2.5 shows how a cGAN is different from an unconditional or simple GAN.

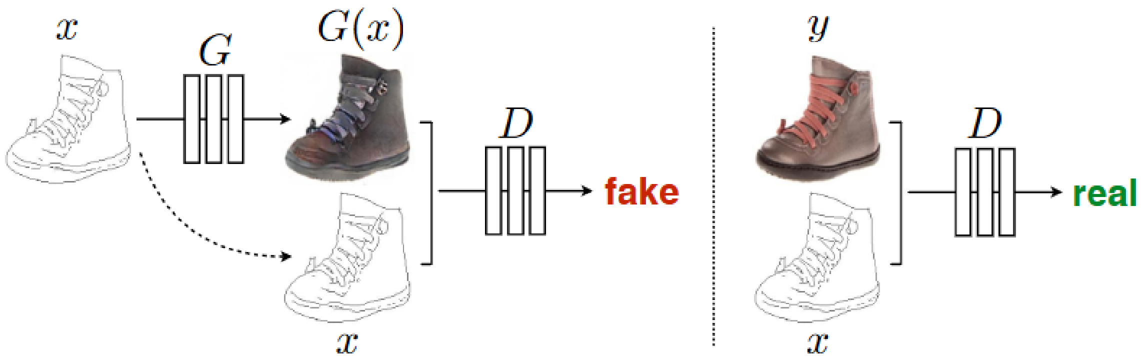


Figure 2: Training a conditional GAN to map edges→photo. The discriminator, D, learns to classify between fake (synthesized by the generator) and real {edge, photo} tuples. The generator, G, learns to fool the discriminator. Unlike an unconditional GAN, both the generator and discriminator observe the input edge map.

Figure 2.5: cGAN Working Mechanism

2.9 Transfer Learning

Given a target learning task T_t based on a dataset D_t, and a source task T_s aided by a larger dataset D_s, transfer learning aims to improve the performance of the predictive function $f_T(\cdot)$ for T_t by discovering and transferring helpful latent knowledge from D_s and T_s, where either the datasets differ ($D_s \neq D_t$), the tasks differ ($T_s \neq T_t$), or both. Typically, the source dataset is much larger than the target dataset, i.e., $N_s \gg N_t$.[25] This is demonstrated in Fig.2.6.

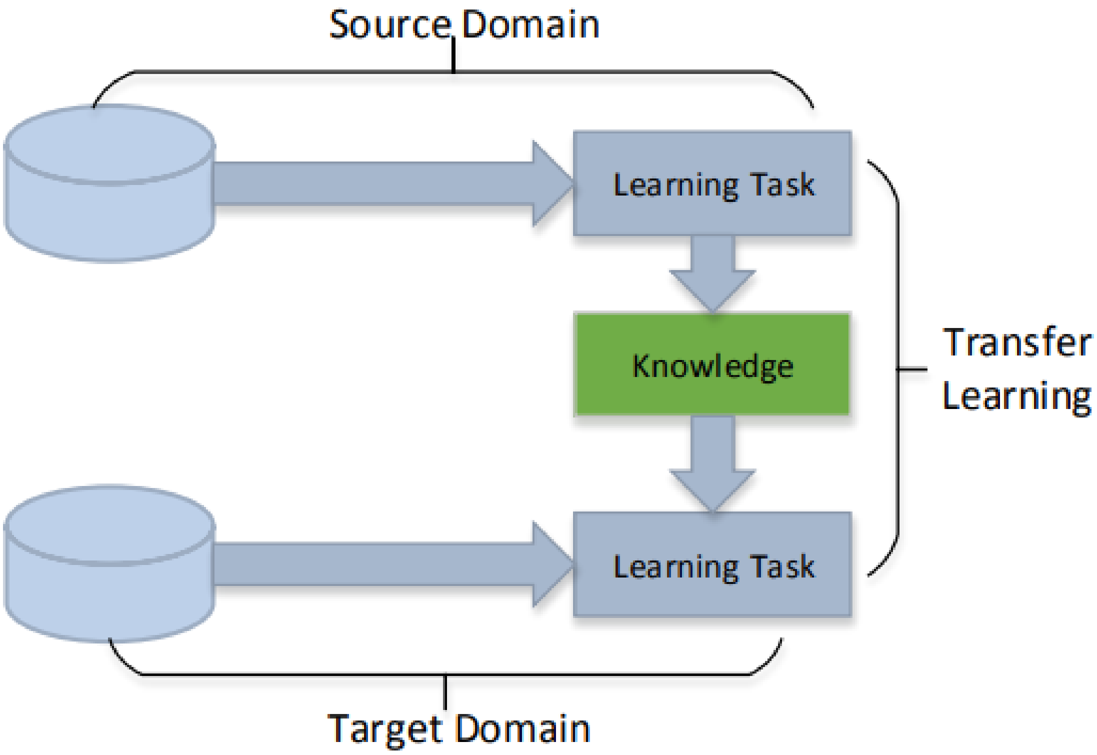


Figure 2.6: Transfer Learning
[25]

2.10 Diffusion Models

Interestingly, diffusion models which are now talk of the town owe their origin to thermodynamics and statistical physics. The process of diffusion in diffusion models draws its inspiration from a simple process such as diffusion of ink droplets through water. In real life, once diffused in water, it is impossible to separate and regain the original ink droplets from water. A diffusion model in Deep Learning (DL) tries to achieve this and recover the original structure of data which has been lost to noise addition.[26]

Diffusion models are a family of generative models that define a Markov chain of successive transformations, gradually converting data into noise (the forward process), and then learning to reverse this process (reverse process) to get structured data out of noise.[26], [27]

2.10.1 Forward Diffusion Process

The forward process adds small amounts of Gaussian noise to the data over T time steps. Let $\mathbf{x}_0 \sim q(\mathbf{x}_0)$ be a data sample from the true data distribution. The process defines a sequence $\mathbf{x}_1, \ldots, \mathbf{x}_T$ such that:

$$q(\mathbf{x}_{1:T} \mid \mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t \mid \mathbf{x}_{t-1}),$$

where each transition is defined as:

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t \mathbf{I}),$$

and $\beta_t \in (0, 1)$ is a variance schedule controlling the noise level at each step. As $t \to T$, $\mathbf{x}_T$ approaches a standard normal distribution $\mathcal{N}(0, \mathbf{I})$.[27]

2.10.2 Reverse Diffusion Process

The goal of the diffusion model is to learn the reverse Markov process:

$$p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t),$$

where $p(\mathbf{x}_T) = \mathcal{N}(0, \mathbf{I})$, and each $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ is modeled as a Gaussian whose mean and variance are predicted by a neural network parameterized by θ. This process denoises $\mathbf{x}_T$ step-by-step to obtain $\mathbf{x}_0$.[27]

2.10.3 Training Objective

The model is trained to predict the noise added in the forward process. The loss function is derived by minimizing the variational bound on the negative log-likelihood:

$$\mathcal{L}_{\text{vlb}} = \mathbb{E}_q \left[\sum_{t=1}^{T} D_{\text{KL}}(q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)) \right],$$

and in practice, a simplified loss is often used:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{\mathbf{x}_0, \boldsymbol{\epsilon}, t} \left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2 \right],$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ is the noise, and $\boldsymbol{\epsilon}_\theta$ is the model's prediction of that noise given the noisy input $\mathbf{x}_t$ and timestep t.[27]

2.11 Fréchet Inception Distance(FID)

Fréchet Inception Distance (FID) is a score suggested by Heusel et al. in 2017 to quantify image quality generated by generative models, for example, Generative Adversarial Networks (GANs). FID approximates the similarity between the real and synthetic image distributions by computing the statistics of the feature representations extracted from a pretrained Inception-v3 network. Specifically, FID computes the Fréchet distance (also known as the Wasserstein-2 distance) between two multivariate Gaussian distributions tuned to the feature representations of real and synthetic images. A lower FID score indicates that the synthetic images are more similar to real images in visual quality and diversity. FID has become a standard evaluation metric for assessing the performance of generative models in image synthesis tasks.[28]

2.12 Redundancy Ratio

Redundancy Ratio is a metric used to measure the degree of duplication or similarity among samples in a dataset. In the context of data augmentation, it quantifies how much repetitive content exists within a set of generated images. A high redundancy ratio indicates that many generated samples are visually or semantically similar, suggesting low diversity. Conversely, a low redundancy ratio implies greater variation among the outputs.

2.13 Low-Rank Adaptation

Low-Rank Adaptation (LoRA)[29] is a low parameter fine-tuning method introduced to adapt large pretrained models to downstream tasks without modifying all the model weights. Instead of modifying the whole weight matrices, LoRA introduces trainable rank-decomposition matrices in every layer without modifying the original weights. The update is specifically designed as a low-rank decomposition of the weight changes as follows:

$W_0 + \Delta W$, where $\Delta W = BA$, with $A \in \mathbb{R}^{r \times d}$ and $B \in \mathbb{R}^{d \times r}$, for a low rank r[29]

This significantly reduces the number of trainable parameters, speeds up training, and reduces memory use—rendering LoRA especially well-adapted to fine-tuning large diffusion or language models for new domains or datasets without full retraining.

2.14 Summary

This chapter was about laying the groundwork for image synthesis tasks using generative Artificial Intelligence (AI) models. It explained the basic principles of DNNs which get modified in various ways for different outcomes. The chapter shed light on the differences and use cases of labeled and unlabeled data. How data augmentation can be leveraged to make up for low data availability was discussed here. The growing importance of AI-based technology for the creation of visual guidance systems was discussed too. Generative models are the latest addition in the list of techniques that help improve the performance of such systems. Deterministic and probabilistic models were introduced. And finally the concepts of cGANs, diffusion models, and their performance evaluation metrics were discussed. The next chapter specifically dives deep into the conceptualization of the experiments that this work is based upon and state-of-the-art methods which have aided in solving the problem addressed in this thesis.

Chapter 3

Generation of Images Using Semantic Segmentation Masks

This chapter explores methods previously used to create synthetic datasets and their effectiveness in terms of producing diverse and realistic images. These earlier approaches provided the foundation and inspiration for this thesis and helped shape the direction and objectives of this work.

3.1 cGANs for Synthetic Image Generation

GANs were originally suggested for the purpose of creating realistic data by learning about the underlying distribution of a dataset. These original GANs, however, did not gain direct control over what kind of data they were generating — they could only create samples after training in an unstructured or latent-dependent way, with little usefulness for particular generation tasks.[22] The introduction of cGANs by Mirza and Osindero (2014) broke this limitation by conditioning both generator and discriminator on an additional input variable. This conditioning — often in the form of segmentation maps, class labels, or other structured data — enabled guided data generation, making it possible to synthesize outputs in alignment with specific labels or constraints.[30]

One of the earliest and most impactful applications of cGANs in the image domain has been in image-to-image translation, where models have learned to convert a structured input (e.g., an edge map or a segmentation mask) into a realistic output image. This was led by the Pix2Pix model, which introduced the utilization of semantic maps as conditioning data.[4] Pix2PixHD, a high-resolution variant of this model, extended this idea even further by using multi-scale discriminators and a coarse-to-fine generator to enable the generation of more detailed and visually coherent images.[5] Pix2PixHD also showed how these labels could be modified to manipulate scenes by changing positions of various objects. This proved the potential of cGANs in expanding datasets with greater diversity. The same paper also demonstrated cGANs' capability to create multiple images with the same input condition.[5] This structural guidance was essential for tasks where spatial layout matters — such as urban scene synthesis, and in the case of this thesis, pedestrian walkway simulation. Pix2PixHD thus became a natural choice for generating photorealistic, semantically valid street-level imagery from sparse annotations.

Later enhancements, such as SPADE (Park et al., 2019), built on Pix2PixHD's capability by enabling better spatial modulation through semantic-aware normalization layers. SPADE allowed for greater variability in outputs while preserving semantic structures better. [6] Despite diffusion models outperforming cGANs in terms of image quality and diversity as shown in [31], diffusion models are computationally expensive making cGANs still a popular choice.[32]

3.2 Diffusion Models for Synthetic Image Generation

Diffusion models although originally inspired by thermodynamics in [26], were refined and reintroduced as Denoising Diffusion Probabilistic Models(DDPMs) in 2020 in [27] for unconditional image generation. Diffusion models outperformed GANs on diversity metric such as FID and could also effectively solve the common issue with GANs which is that of mode collapse.[31]. To gain more control over image synthesis through image class/label, [33] explicitly introduced classifier-guided diffusion, enabling class-conditional image generation. It conditioned the reverse denoising process using the gradients of a pretrained classifier (e.g., on ImageNet). It was the first to demonstrate high-quality conditional image synthesis using diffusion.[34].

Diffusion models conditioned on semantic segmentation masks and/or general purpose image-to-image translation-capable diffusion were introduced in 2022 in works like [35],[8]. While Palette[35] did prove to be successful in image translation tasks such as inpainting, colorization, uncropping and JPEG restoration, its applicability was not directly useful in semantic segmentation masks being used as inputs for image generation. [34] introduced the idea of performing diffusion in a compressed latent space, drastically reducing computational cost. Latent Diffusion Models (LDMs) formed the basis of later work in stable diffusion models. Stable diffusion was originally developed as a text-to-image model. It is a latent diffusion architecture-based model, which allows it to generate high-quality images from text prompts efficiently. This was a revolutionary development in generative AI, as users were now able to create detailed images from descriptive text inputs. Subsequently, the ability of Stable Diffusion was extended to semantic map-to-image translation. This was achieved via the addition of ControlNet[7], a neural network design for providing further control to image synthesis by inserting various types of conditioning inputs, including segmentation maps. ControlNet enables high-resolution control over the synthesis of images from an input that can closely track the structure provided by semantic maps.

Briefly, Stable Diffusion was initially a text-to-image model that was later applied to semantic map-to-image translation by adding ControlNet in order to further its application in structured image synthesis tasks.

In [8], semantic segmentation maps were used to guide the diffusion model for generating photorealistic images in a structure-preserving manner, showing that semantic-to-image synthesis via diffusion could surpass GAN-based methods in terms of realism and diversity.

Both [7] and [8] emerged as practical choices, offering pretrained models that could be adapted for structure-conditioned generation. The subsequent introduction of ControlNet extended Stable Diffusion to accept segmentation masks as control inputs, providing a lightweight yet effective path toward implementing the SDM-style synthesis pipeline. As such, the literature gradually shaped the decision to adopt Stable Diffusion + ControlNet in this thesis, balancing semantic guidance with computational feasibility.

3.3 Transfer Learning in Generative Models

Transfer learning has become a desirable method in generative modeling, particularly when it comes to application in those areas where there are no enormous quantities of labeled data. For GANs, it is usually applied by fine-tuning previously trained models such as Pix2Pix or Pix2PixHD that were initially trained using large datasets such as Cityscapes.[5][1] Pretrained models can then be adapted to similar but data-constrained areas with very little retraining. For instance, the IDD-Pix2Pix project shows how a Pix2PixHD model trained initially on road scenes was reused to create images for Indian traffic environments.[36]

For diffusion models, full fine-tuning is usually avoided owing to heavy computational overhead. Rather, light adaptation techniques such as DreamBooth[37], Textual Inversion[38], and LoRA[29] have emerged. These techniques allow new styles, objects, or subjects to be added to models like Stable Diffusion without adjusting the underlying weights. Stable Diffusion itself is a variation of the LDM[34], pre-trained on the vast LAION-5B dataset[39], and has been employed as a base for many downstream applications. ControlNet is also a form of transfer learning where a secondary branch of the neural network is trained for structure-based conditioning and the base diffusion model is kept frozen. The modular architecture facilitates domain adaptation with fine-grained control, and transfer learning is thereby an effective and useful tool for generative model deployment in data-limited regimes.[7]

3.4 Summary

This chapter covered the most major generative models specifically, cGANs and diffusion models that formed the basis of this study's experiments. Since transfer learning is an important aspect of image generation in both the aforementioned methods under comparative study in this thesis, its realistic implementations were all highlighted. The next chapter will deal with foundational setup for the work conducted in this thesis. That will include dataset preparation, architecture and preprocessing.

Chapter 4

Methodological Foundations: Dataset, Preprocessing, and Architecture

The methodology presented in this thesis integrates detailed data preparation, training of cGANs both from scratch and via transfer learning, and fine-tuning of Stable Diffusion with ControlNet. These approaches were employed to expand the SENSATION dataset by generating new photorealistic images from segmentation masks while preserving their underlying semantic structures.

4.1 Dataset Preparation

Major efforts were dedicated to data preprocessing, a process highlighted as a key improvement area in previous works such as [5]. The preprocessing pipeline included initial dataset analysis and visualization, cleaning, and a two-step augmentation process aimed at class balancing and diversity enhancement. A redundancy ratio check was also applied to assess the contribution of the augmented data.

4.1.1 The SENSATION Dataset

Unlike existing datasets such as Cityscapes[1], SENSATION ensures that sidewalks are consistently centered in the frame — a critical requirement for forward-facing navigation guidance systems for the visually impaired. Thus, the final expansion was to be made to this dataset. This dataset is a combination of images from three different sources, resulting in a total of 3,233 images with corresponding masks. The sources are as follows:

1. Mapillary Dataset: Consists of 1,000 images with corresponding masks, selected where the sidewalk is predominantly centered. 2. Sidewalk Dataset from Hugging Face: Comprises 1,000 images with corresponding masks. 3. Custom Dataset: Created and labeled by the Pattern Recognition Lab Team at Fredrich Alexander University (FAU) Erlangen, including 1,233 images with corresponding masks. Mapillary and Hugging Face datasets are open-source, while the custom dataset was used with permission from the FAU Pattern Recognition Lab.

Fig.4.1 shows an example image and its corresponding RGB mask. Even though the figure shows an RGB mask, the ones used in model training and testing are their grayscale versions where each pixel value bears the class ID to which it belongs. The images were all high resolution but with varying resolutions which were later reshaped before being used in different models as per the models' input requirements. The average resolution of an image though was 2048×1024.

The masks were created post semantic segmentation of the images thus, creating a labeled dataset with 3,233 image-mask pairs. There are 12 classes within data labels as can be seen in Table4.1.

Class	**Class ID**
0	Background
1	Road
2	Sidewalk
3	Bikelane
4	Person
5	Car
6	Bicycle
7	Traffic Sign(front)
8	Traffic Light
9	Obstacle
10	Stairs
11	Crosswalk

Table 4.1: Classes and Class IDs

4.1.2 Data Visualization

The idea behind data visualization was to gain a closer look at the distribution and representation of classes within the dataset. This was necessary to ensure that class-

Figure 4.1: Example Image and Corresponding Label Map from the SENSATION Dataset

imbalance would not interfere with training quality, Data visualization was also needed to find the direction in which augmentation had to proceed.

The objective of data visualization was to gain a comprehensive understanding of the class distribution within the SENSATION dataset and to identify imbalance patterns that could influence model performance. The visualization process was approached in a layered fashion — beginning with broader summaries and gradually moving toward finer, class-wise relationships.

A bar plot(shown in Fig.4.2) was the most natural choice used to visualize the frequency of each class label across the dataset. This directly pointed at the imbalance, showing that certain classes such as person, traffic light, and traffic sign were significantly underrepresented compared to dominant classes like road and sidewalk.

To explore inter-class relationships and co-occurrence, a stacked bar plot was then generated to observe how multiple classes appeared together within images. However, this visualization proved less effective for understanding overlap dynamics, especially for the rare classes.

This led to the generation of a correlation heatmap, initially including all classes. It was quickly observed that the global heatmap was skewed by class imbalance, making correlations appear stronger than they were due to large sample disparities. To correct this, a refined correlation matrix which can be seen in Fig.4.3, was computed using only the underrepresented classes.

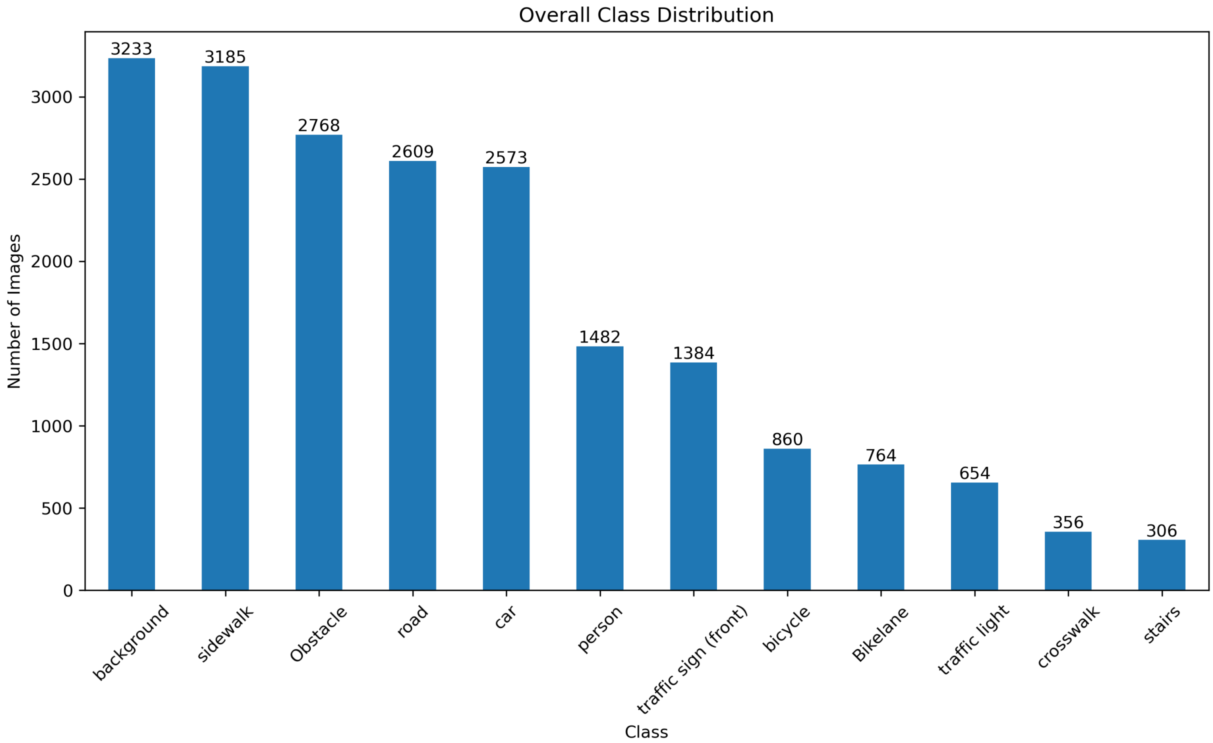


Figure 4.2: Class Distribution within the SENSATION Dataset

This focused heatmap revealed that correlations between underrepresented classes were generally low — around 0.3 to 0.4 between classes such as traffic sign, traffic light, and person. This low co-occurrence suggested that these classes could be safely augmented independently without introducing excessive redundancy or semantic conflict in the dataset.

To guide augmentation goals, a target threshold of 2500 images per class was set. This number was chosen heuristically — it sits near the mean count of well-represented classes (2700), ensuring enough representation without bloating the dataset unnecessarily.

Together, these visualizations not only confirmed the presence of class imbalance but also informed a data-driven augmentation strategy designed to increase diversity without compromising semantic coherence.

4.1.3 Data Augmentation

This part of data preprocessing was carried out to ensure that the main models, that is, cGANs and diffusion received quantity-wise more data that the SENSATION dataset could originally provide. Traditional augmentation techniques were applied to the images and masks to expand the dataset. It was carried out in a two-step pipeline designed to increase

Figure 4.3: Correlation Heatmap of Underrepresented Classes

dataset diversity and correct class imbalance, especially for underrepresented classes such as crosswalk, traffic light, and person. The augmentation workflow was built iteratively, with logic driven by earlier visualizations of class distribution and correlation.

Overall Strategy and Planning

The augmentation pipeline mainly consisted of 2 steps to leverage the use of selective cropping and geometric transformations so as to produce as many varying images using limited number of masks. The same augmentation was applied at every stage to the image and its corresponding mask only from the under-represented classes which were namely, *bicycle*, *bikelane*, *crosswalk*, *person*, *stairs*, *traffic light* and *traffic sign*.

Step 1: Cropping Based on Label ID

The first stage of augmentation focused on generating additional samples for underrepresented classes by cropping subregions from images in which these classes were present. This was accomplished using a custom Python script, which automated the detection and cropping of bounding boxes corresponding to selected label IDs in the semantic masks.

The script was highly configurable. It allowed specification of a threshold parameter to control the expansion margin around the bounding box, and accepted a label list defining which classes to crop. This enabled precise extraction of meaningful regions without excluding necessary contextual information.

To diversify the training data and avoid repetitive patterns in the cropped outputs, a randomization strategy was introduced. Specifically, each cropping operation was guided by a combination of three label IDs. The first label ID in the list was always fixed to be one of the underrepresented classes (e.g., person, traffic light, crosswalk), while the second and third label IDs were randomly shuffled from a pool of remaining class labels. This ensured that every crop focused on boosting minority class representation, while still incorporating natural variability from other co-occurring objects in the scene.

Step 2: Geometric Transformations

Following class-specific semantic cropping, the second stage of augmentation focused on introducing geometric variability to further enhance the diversity of the training data. This was achieved using the Albumentations library, which offers a wide range of fast and flexible image transformation functions suitable for paired image-mask augmentation.

Among the transformations explored, horizontal flipping proved to be the most effective and semantically safe technique. It introduced meaningful spatial variation without compromising the integrity of the segmentation masks. Other geometric transformations such as rotation, scaling, and translation were also tested; however, their utility was limited, and some were found to degrade the contextual relevance of the image content, particularly in structured urban scenes.

In addition, certain photometric transformations like brightness adjustment, contrast modification, and color jitter were intentionally omitted or reconsidered. Since the semantic masks are often represented in grayscale or categorical RGB formats, these transformations had negligible impact or risked introducing inconsistency between image-mask pairs.

Crucially, these augmentations were applied only to images containing underrepresented classes, as identified during the visualization phase. This selective strategy ensured that the data augmentation process improved class balance without further inflating already dominant categories.

The augmentation pipeline was configured to operate in batch mode, enabling efficient processing and automated saving of augmented image-mask pairs. This setup supported streamlined dataset expansion and ensured consistent formatting for subsequent model training.

4.1.4 Special Label Remapping for Transfer Learning

For the fine-tuning of cGAN model, Pix2PixHD[5] which was already trained on the Cityscapes[1] dataset, it was necessary to align the class IDs of the SENSATION dataset as per the class IDs of the Cityscapes dataset for the common classes existing in both the datasets. For the uncommon classes, an approximation like the class '*background*' in the SENSATION dataset being mapped to '*vegetation*' in the Cityscapes dataset, was used.

4.2 Architectural Details

The 3 main models used in this thesis were Pix2PixHD[5], Semantic Diffusion Model (SDM) and Stable Diffusion combined with ControlNet[7], [34] as shown in Fig.4.4.

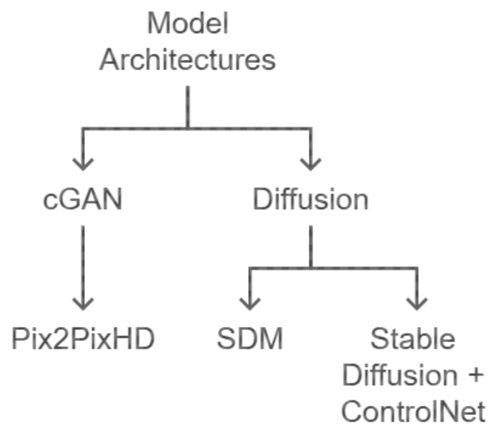


Figure 4.4: Major Model Architectures

4.2.1 Pix2PixHD

Pix2PixHD is essentially a cGAN. Thus, before understanding its architecture it is important to understand the architecture of a cGAN as originally proposed in [30].

Architecture of a Conditional Generative Adversarial Network

cGANs extend the standard GAN framework by introducing conditioning information $\mathbf{y}$ into both the generator and the discriminator as shown in Fig.4.5. This conditioning variable can represent class labels, segmentation masks, or any structured data that provides semantic guidance. The generator G takes a noise vector $\mathbf{z} \sim p_z(\mathbf{z})$ and the condition $\mathbf{y}$ to produce a synthetic output $G(\mathbf{z} \mid \mathbf{y})$. This allows the generator to produce outputs that are not only realistic but also aligned with the input condition.

The discriminator D receives both an image $\mathbf{x}$ and the conditioning variable $\mathbf{y}$, and its role is to assess whether the image is real and whether it matches the given condition. This joint input enables the discriminator to learn structure- or class-aware features, making it a more effective evaluator in tasks that require semantic consistency.

In most implementations, the conditioning vector $\mathbf{y}$ is concatenated with the noise vector $\mathbf{z}$ in the generator and with the image $\mathbf{x}$ in the discriminator. This design allows for flexible incorporation of various modalities and has proven particularly useful in tasks like semantic image synthesis and domain translation, where control over the generated output is essential.

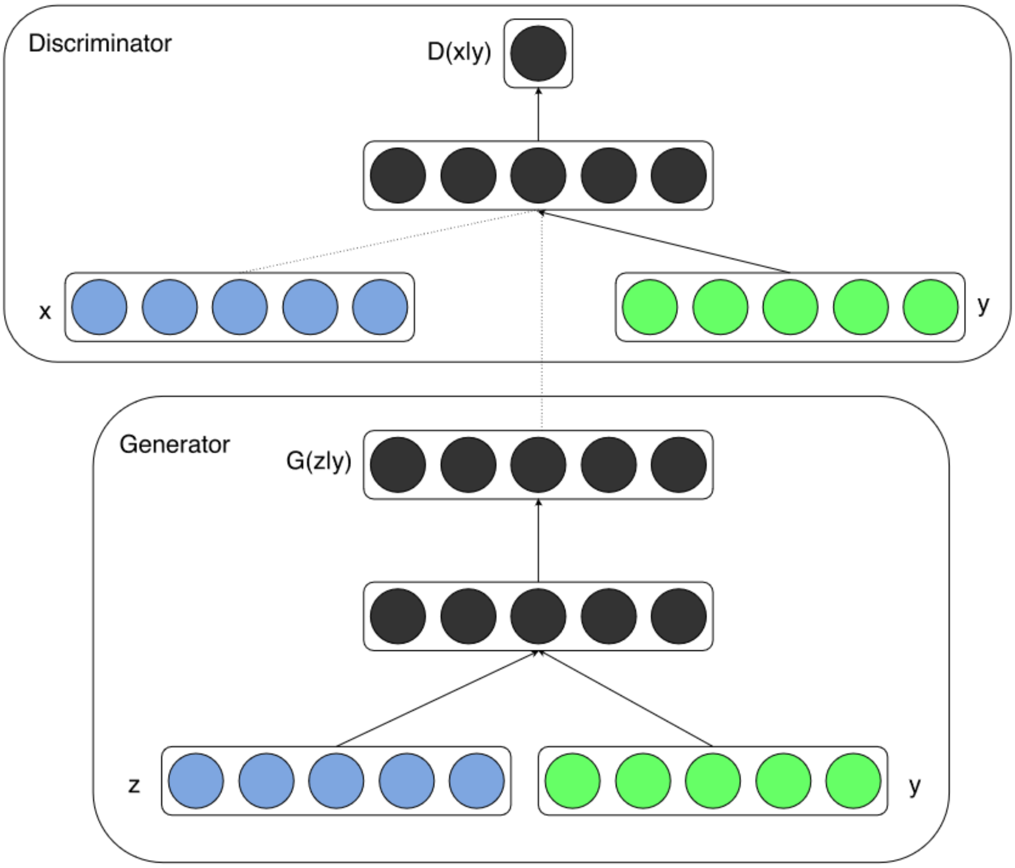


Figure 4.5: Conditional Adversarial Generative Net
[30]

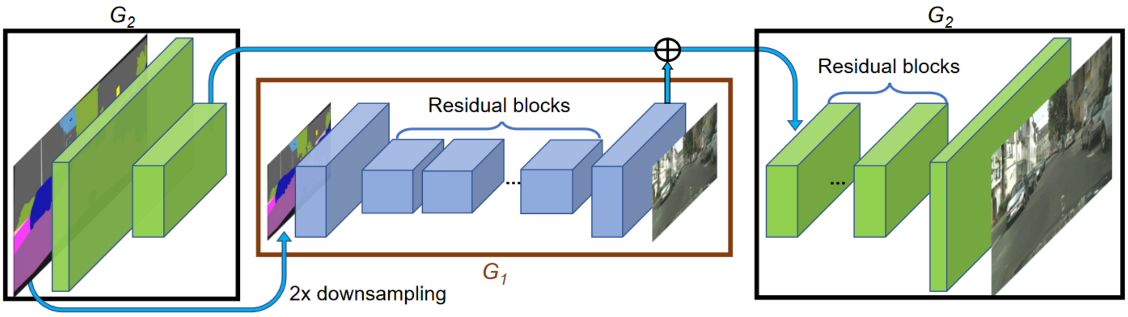


Figure 4.6: Generator Architecture of Pix2PixHD
[5]

Architecture of Pix2PixHD

Pix2PixHd is an improvisation of the base model, pix2pix[4] with the introduction of a coarse-to-fine generator as shown in Fig.4.6 and a multi-scale discriminator architecture. Its generator is a two-stage coarse-to-fine network: a global generator that produces a low-resolution output, and a local enhancer that generates higher resolution by adding fine-grained details to it. Both stages include a convolutional front-end, a stack of residual blocks, and a transposed convolutional decoder. To compose global and local context, the global generator and local enhancer features are merged during generation. The modular design allows for scalable image synthesis, with even larger resolutions being accommodated by incorporating additional enhancers. This design has been utilized greatly in the thesis when carrying out transfer learning.

To preserve visual fidelity at varying spatial scales, Pix2PixHD employs three multi-scale discriminators at full, half, and quarter resolutions. Each discriminator shares the same architecture but examines image realism at different scales of detail. The coarse-scale discriminators enforce global structural consistency, whereas the fine-scale ones teach the generator to produce crisp textures. This multi-scale strategy not only improves the output quality but also renders the training process efficient since extension to a higher resolution simply requires adding another discriminator. Together, the hierarchical generator and multi-scale discriminators make Pix2PixHD highly effective for high-resolution tasks such as semantic-to-image translation.[5]

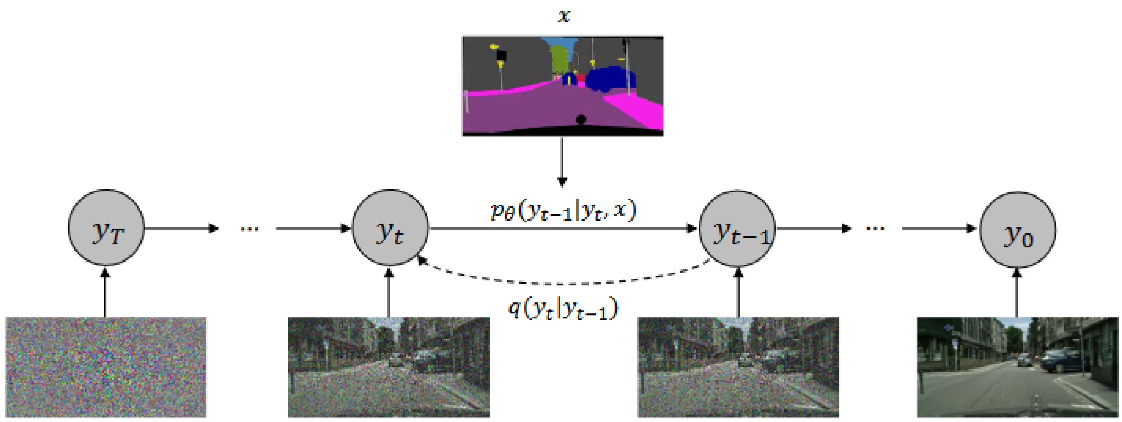


Figure 4.7: Semantic Label Input in SDM [8]

4.2.2 Semantic Diffusion Model

SDM's uniqueness is attributed to the way it feeds semantic map and noisy image to its architecture. Other conditional diffusion models like Palette[35] directly feed the semantic map and noisy image as input to a U-Net structure while SDM feeds noisy image to the encoder of the U-Net structure while the semantic layout to the decoder by multi-layer spatially-adaptive normalization operators as shown in Fig.4.7.[8] The noisy image is passed through the encoder of the denoising network, while the semantic layout is integrated into the decoder using spatially-adaptive normalization layers applied across multiple levels.

Encoder

The encoder processes the noisy input image using a series of *semantic diffusion encoder residual blocks* (SDEResblocks) and self-attention layers. Each SDEResblock includes a convolutional layer, the SiLU activation function (defined as $f(x) = x \cdot \text{sigmoid}(x)$), and group normalization. To incorporate timestep information t, each block applies learnable scaling and shifting using parameters $w(t) \in \mathbb{R}^{1\times1\times C}$ and $b(t) \in \mathbb{R}^{1\times1\times C}$, modifying intermediate activations as $f^{i+1} = w(t) \cdot f^i + b(t)$. Additionally, attention blocks—based on self-attention with 1×1 convolutions—are applied at selected resolutions (32×32, 16×16, and 8×8), capturing long-range dependencies and improving semantic encoding of the noisy latent.

Decoder

In the decoder, SDM injects semantic layout information via *semantic diffusion decoder residual blocks* (SDDResblocks), which differ from the encoder by incorporating *spatially-adaptive normalization* (SPADE)[6] instead of group normalization. SPADE modulates the normalized feature maps with learned, spatially-varying scale $\gamma_i(x)$ and bias $\beta_i(x)$ derived from the semantic label map, as formulated by $f^{i+1} = \gamma_i(x) \cdot \mathrm{Norm}(f^i) + \beta_i(x)$. This mechanism allows semantic information to guide denoising at multiple layers in a context-aware manner. The decoder also integrates skip-connections, attention layers, and timestep embeddings, enabling it to generate semantically aligned and temporally consistent outputs through the diffusion process.

4.2.3 Stable Diffusion with ControlNet

Stable Diffusion is an LDM that generates high-quality images by operating in a compressed latent space rather than pixel space. It consists of three key components: a VAE that encodes and decodes images into latent representations, a UNet-based denoising network that performs the diffusion process, and a conditioning module that injects guidance—commonly in the form of text embeddings. This design significantly reduces computational overhead while maintaining photorealistic generation quality.[34]

ControlNet builds on top of this framework to allow structured conditioning using visual inputs such as edge maps, depth maps, or segmentation masks. Rather than altering the pre-trained diffusion model directly, ControlNet replicates the architecture of the UNet and adds zero-initialized convolutional layers—known as zero-convs—that learn to inject the control signal into various layers of the diffusion process. These control features are fused with the base model's latent features at multiple resolutions, enabling fine-grained alignment with the conditioning input while preserving the generative capacity of the original Stable Diffusion network.[7]

In Fig.4.8, the left half of the figure shows the original frozen UNet used in Stable Diffusion, while the right half represents the parallel, trainable copy introduced by ControlNet. The structural condition (e.g., segmentation map) is processed through a series of zero-initialized convolution layers and injected at multiple points throughout the network. These trainable copies enable the model to incorporate spatial guidance without modifying the pre-trained weights of the base model, thus preserving generative quality while achieving fine-grained control over output structure.

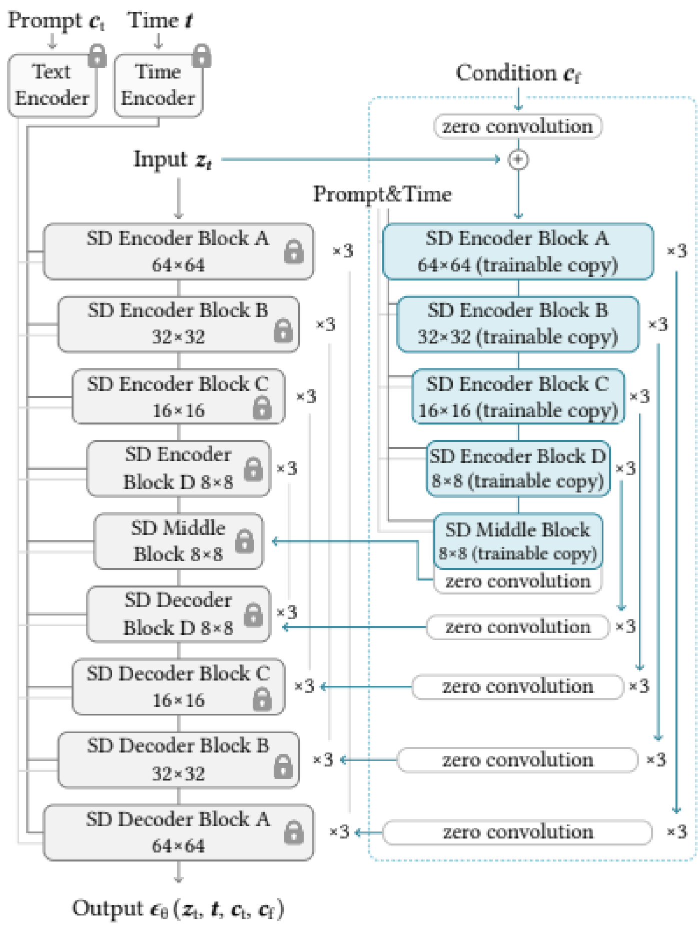


Figure 4.8: Architecture of ControlNet Integrated with Stable Diffusion Model [7]

4.3 Hyperparameters

This section outlines the key hyperparameters that significantly influenced the training and generation performance of the two main models used in this thesis: Pix2PixHD and Stable Diffusion with ControlNet. Understanding and adjusting these hyperparameters was critical in optimizing the models for photorealistic, semantically accurate outputs tailored to the SENSATION dataset.

4.3.1 Hyperparameters Crucial to Pix2PixHD

The performance of the Pix2PixHD model was influenced by a number of architectural and training hyperparameters:

- **Learning Rate (`lr`)**: The base learning rate was set to 0.0001 for stable convergence.
- **Number of Discriminators (`num_D`)**: The model was experimented with both 2 and 3 discriminators to evaluate their effect on image detail and training stability.
- **Number of Epochs**: Training was conducted with different epoch configurations (e.g., `niter` = 50, `niter_decay` = 50), where `niter_fix_global` controlled how long the global generator remained frozen.
- **Generator Filters (`ngf`)**: This was set to 32, affecting the width of the generator's convolutional layers.
- **Generator Freezing (`niter_fix_global`)**: To prevent catastrophic forgetting when fine-tuning, the global generator layers were frozen for the initial epochs. Experiments were conducted with values like 10 and 40, with 40 yielding the most stable and realistic outputs.

4.3.2 Hyperparameters Crucial to Stable Diffusion and ControlNet Model

The diffusion-based pipeline required tuning of both diffusion and ControlNet-specific parameters:

- **Guidance Scale (`guidance_scale`)**: This controls how strongly the model follows the text prompt.

- **ControlNet Conditioning Scale (`controlnet_conditioning_scale`)**: This adjusts the strength of structural guidance from input masks.
- **Strength (`strength`)**: Determines how much the initial noise or input image is preserved during generation.
- **LoRA Fine-tuning Parameters**: These included learning rate and number of steps, optimized to adapt the pretrained model efficiently to SENSATION.

4.4 Summary

This chapter outlined methodological underpinnings on which this thesis stands, beginning with an introduction of the SENSATION dataset used to carry out semantic-to-image synthesis. The more than 3,000 image-mask pairs from multiple sources that constitute the dataset were investigated by purposeful visualizations to investigate class imbalance as well as inter-class relations. Following this insight, two-stage data augmentation strategy was planned. This entailed selective cropping of minority classes and subsequent geometric transformations to increase diversity without loss of semantic meaning.

The chapter also introduced the key architectures used in this research—Pix2PixHD, Stable Diffusion, and ControlNet—each selected based on their potential in photorealistic image synthesis under structural constraint. Pix2PixHD was presented as a coarse-to-fine conditional GAN with multi-scale discriminators, while Stable Diffusion operates in a latent space to enable efficient high-resolution synthesis. ControlNet, built on top of Stable Diffusion, allows for fine-grained structural guidance through the application of zero-convolution modules and spatially-aware feature merging. Both models form the base of the image generation pipeline researched in the later experimental tests. The next chapter will be extensively focused on various experiments that were carried out in a planned and logically-driven sequence.

Chapter 5

Experimentation

Experimental setup comprised of the entire machine learning pipeline for different architectures that were selected and mentioned in the previous chapter. Since the objective of the study was to create diverse and good quality images while preserving the semantic structures of the masks in the SENSATION dataset, the experiments conducted generally proceeded in the following steps:

(1) Testing if the selected architecture is functional and gives results as shown in its research paper with one of the datasets used in its original research paper.

(2) Modifying the SENSATION dataset as per the input requirement of the model.

(3) Training or fine-tuning the model with the SENSATION dataset.

(4) Hyperparameter tuning for the improvement of results.

(5) Repeat steps 2 to 4 with the augmented dataset which was achieved as a part of data preprocessing as mentioned in the previous chapter.

(6) Check if the results have improved with the augmented dataset.

(7) Move on to the next model for experimentation.

All of the experiments were conducted with the HPC-cluster at FAU. Table 5.1 further shows the main system specifications that were used in the experiments. Table 5.2 shows main libraries used for various purposes in the experiments.

Table 5.1: System Specifications

Component	Specification
Operating System	Linux
Programming Language	Python 3.9
Deep Learning Framework	PyTorch 2.5.1
CUDA Version	12.9
GPU	NVIDIA A100-SXM4-40GB
Environment Manager	Anaconda
Experiment Tracking	MLflow
Cluster Scheduler	SLURM

Table 5.2: Libraries and Tools Used

Category	Libraries / Tools
Deep Learning	PyTorch, Torchvision, Hugging Face Diffusers
Data Augmentation	Albumentations
Experiment Tracking	TensorBoard, MLflow
Image Processing	OpenCV, PIL (Python Imaging Library)
Data Manipulation	NumPy, Pandas
Visualization	Matplotlib, Seaborn

5.1 Experiment 1: Pix2PixHD

The experiments that were carried out with the Pix2PixHD model were broadly: 1. Fine-tuning the model trained on the Cityscapes Dataset 2. Training the model on the augmented SENSATION Dataset This implied training, hyperparameter tuning, testing and gradual improvisation of the methods as already mentioned.

5.1.1 Fine-tuning Model Trained on Cityscapes

[40] provides the official implementation of the Pix2PixHd model[5] and it also provides the weights of a model trained on the Cityscapes dataset. Despite the fact that the Cityscapes dataset does not contain images that are focused on pedestrian navigation, it does provide a good overlap with the required scenes, that is, urban city scenes with classes such as roads, sidewalks, traffic signs, traffic lights etc.[1] As a result, fine-tuning a model which is already trained on the Cityscapes dataset serves as a good starting point. As per the already mentioned workflow, the very first step was to verify the functionality of this model

with test images from the Cityscapes dataset. The test results were a replica of the ones mentioned in the paper[5] thus, establishing its use in further experimentation.

Since Pix2PixHD relies on accurate semantic segmentation masks as conditions for image synthesis, it became necessary to address potential misalignment in label IDs and color encodings. The Cityscapes dataset employs a set of 35 class labels, while SENSATION uses 12 classes with a distinct ID schema. As a result, a custom Python script was implemented to remap SENSATION's labels to Cityscapes-compatible categories. In this process, seven of the SENSATION classes were found to correspond directly with classes in Cityscapes. The remaining five were approximated using semantically similar Cityscapes classes,such as mapping *background* to *vegetation* or *obstacle* to *wall* or *fence*.

Two separate remapping strategies were developed:

- **Remapping 1:** Direct class mappings wherever possible, with less emphasis on how background and undefined classes were treated.

- **Remapping 2:** A refined approach where `background` was explicitly mapped to `vegetation` to better align with the Cityscapes model's learned features.

Discriminator Configuration and Remapping Experiments

Following the initial remapping, the model was trained using the first remapping configuration with the default three discriminators (`-num_D 3`). This training was conducted for 100 epochs with a learning rate of 0.0001. Although the generator loss (`loss_G`) decreased to about 16.5 by the 90th epoch, the discriminator loss (`loss_D`) quickly approached zero. This indicated that the discriminator was overpowering the generator, a known issue in adversarial training that often leads to mode collapse or suboptimal image synthesis.

To address this, a follow-up experiment reduced the number of PatchGAN discriminators from three to two (`-num_D 2`) in an attempt to rebalance the adversarial dynamics. However, this configuration did not result in any noticeable change in the learning curves or the quality of generated images. The training loss plateaued, and image synthesis remained unsatisfactory.

In response, the training was repeated with the second remapping strategy (background mapped to vegetation) while reverting to three discriminators (`-num_D 3`). This combination immediately yielded better results in terms of training. Even within the first epoch, the synthetic images exhibited stronger structural integrity and more realistic textures than previous runs. The discriminator loss (`loss_D`) stabilized around 0.55—compared to 0.35

in the previous experiment—suggesting a healthier balance between the generator and discriminator.

Freezing Strategy and Observations

With the second remapping and three discriminators proving most effective so far, the next variable examined was the generator's layer freezing strategy. Pix2PixHD allows partial freezing of the global generator network via the `-niter_fix_global` parameter. Initially, the global layers were frozen for 10 epochs, which helped retain pretrained features while allowing the local enhancer to adapt to the new data.

A final round of experimentation increased this freeze duration to 20 epochs and then 40 epochs, aiming to extend the stabilization phase and encourage better early learning from the local enhancer. During this phase, it was observed that stabilization of the discriminator loss was more critical than a rapid drop in generator loss. Experiments demonstrated that balanced, gradually decreasing losses resulted in more coherent outputs than aggressively minimized but unstable ones. The experimental and logical flow can be summarized in fig.5.1.

5.1.2 Training Pix2PixHD on the SENSATION Dataset

Since the concept of transfer learning did not seem to offer much in terms of realistic results for the Pix2PixHD model, the next major experiment conducted was with training the Pix2PixHD on the augmented SENSATION dataset. The photorealism observed within the first 20 epochs itself was better than that noticed in any of the transfer learning experiments mentioned in *Experiment 1*. The discriminator loss did not quickly collapse to zero in this case. It stabilized around 0.5 after 10,000 steps and remained so until the training ended as shown in 5.2.This was especially important because the stabilization of discriminator loss could not be achieved easily in transfer learning experiments. The training was conducted for 100 epochs with a learning rate of 0.0001. During the training progress, the quality of synthesized image can be see in Fig.5.3. The model was then subjected to testing.

5.2 Experiment 2: Semantic Diffusion Model

[41] provides the official implementation of the SDM. It also provides weights of a pretrained model, trained on the ADE20k dataset[42]. This dataset contains densely annotated

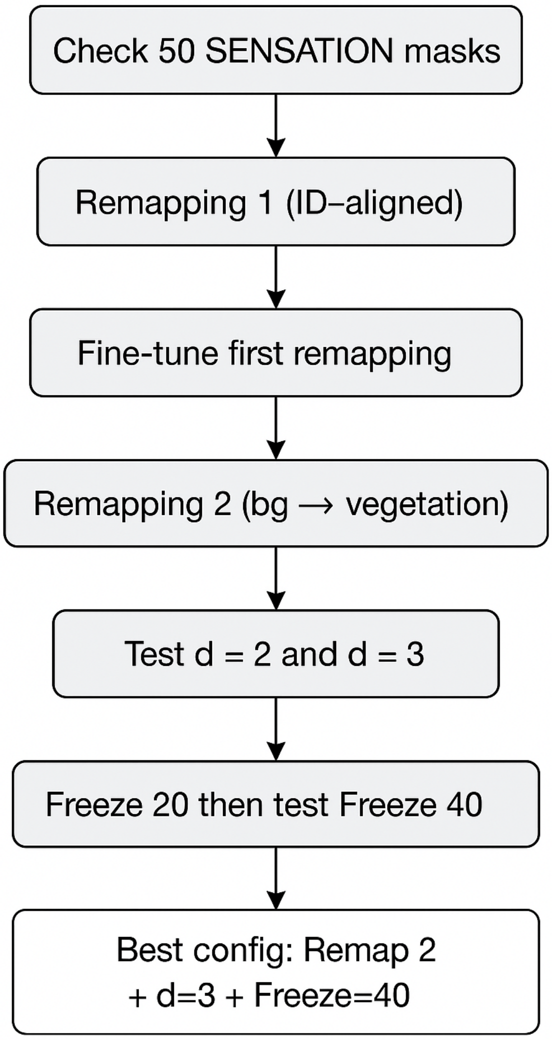


Figure 5.1: Experimental Flow of Fine-tuning

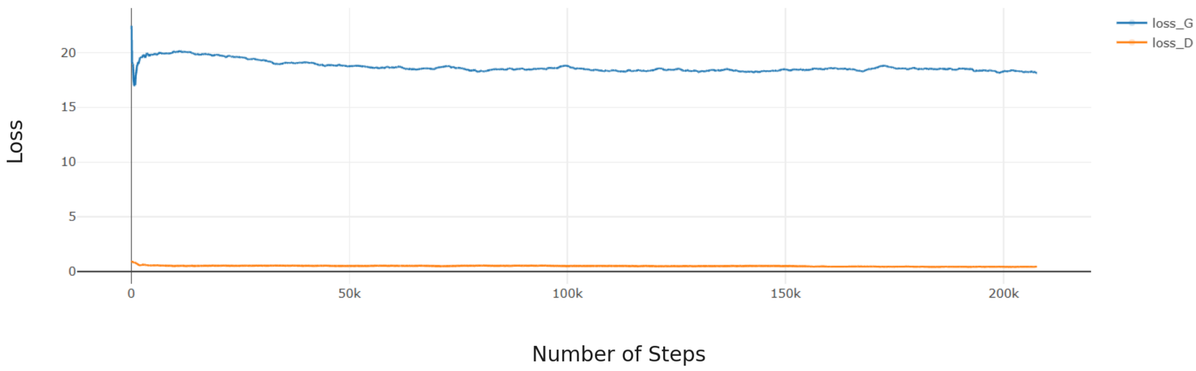


Figure 5.2: Training Loss for Experiment 2

Figure 5.3: Semantic Mask(leftmost), Synthesized Image(middle), Original Image(rightmost) at Epoch 63

everyday life scenes and the model trained on it was considered due to overlap of *urban* scenes' subset with the requirement. Diffusion models are generally used in this manner, i.e., models trained on large generic datasets are often fine-tuned for specific tasks. The process began by flattening the directory structure of the urban subset of SENSATION to match SDM's expected input format. The model was then tested with test images from the ADE20k dataset, urban scenes subset. The produced images were unrealistic in nature, indicating the need to move to a better generic pretrained diffusion model.

5.3 Experiment 3: Stable Diffusion Model with ControlNet

The third experimental setup explored the capabilities of latent diffusion models—specifically, the Stable Diffusion models v1.4 and v1.5 in combination with ControlNet. Diffusers library is an effective way to use the trained model cards from the Huggingface website[43]. The initial part experiments were conducted with an intention to test the functionality of the model as performed with all other models in *Experiment 1, 2*.

Stable Diffusion model is originally a text-to-image generative model which has been adapted using ControlNet which provides an additional control in the form of semantic label. Stable diffusion v1.4 was simply tested to create urban scenes using only text prompts. This generated generic urban city scenes. The aim of this thesis, however, was to generate images with the semantic masks of the SENSATION dataset. The early results also revealed significant shortcomings in realism, particularly when grayscale semantic maps were used as inputs. This prompted a pivot to experimenting with colored masks and even intermediate blurry outputs from the Pix2PixHD model as inputs to the diffusion pipeline, leveraging their visual uncertainty to potentially boost the generative creativity of the model.

In order to achieve generation through semantic masks, Stable Diffusion model v1.5 and ControlNet model were used. This implied the involvement of text prompt and semantic mask for the purpose of synthetic image generation. Thus, the next phase of experiments involved careful trials of:

(1) Prompts for maximizing photorealism and image quality.

(2) Strength factor which assigns weightage to the guidance provided by the semantic mask.

(3) Guidance Scale which assigns weightage to the text prompt in the generation process.

After multiple trials the most realistic and clear images were generated with the following values of the above stated factors:

(1) Text Prompt: "A bustling urban street in Europe as seen by a pedestrian with cars, traffic lights, pedestrians, blue sky, photorealistic, 8k, DSLR photo"

(2) Guidance Scale **8.5**

(3) ControlNet Conditioning Scale **1.2**

The ControlNet model has been used to generate images through the guidance of sketches, edges and other kinds of textural patterns.[7] This implied that blurry images produced as a outputs from the fine-tuning applied to Pix2PixHD could be used as inputs to the ControlNet model. This hypothesis was subjected to experimentation with the same text-prompt, strength and guidance scale factors as stated above.

To adapt the Stable Diffusion + ControlNet pipeline more suitably to the content and structure of the SENSATION dataset, the model was fine-tuned on 2,899 image–colored mask pairs. Instead of retraining from scratch or reloading all weights, the LoRA (Low-Rank Adaptation) method was used to fine-tune the ControlNet model in an efficient way. This process allowed task-specific knowledge to be injected with virtually no computational expense while leaving the original model weights largely intact. Input masks were converted to colored semantic maps, retaining label integrity and enabling richer visual conditioning. Such inputs were important in that they helped preserve the capacity of the model to learn and replicate the specific spatial arrangements relevant to the dataset, especially for pedestrian navigation-focused scenes.

Fine-tuning was done through LoRA using the pre-trained Stable Diffusion backbone and a separately trained ControlNet adapter on SENSATION dataset. Critical hyperparameters like learning rate, conditioning scale factor, and training steps were properly tuned such that tradeoff between semantics preservation and imaginative diversity could be achieved. Through the utilization of LoRA, the model could be allowed to specialize quickly on visual semantics on SENSATION while avoiding overfitting. Qualitative results after fine-tuning showed visibly improved realism, sharper semantic boundaries, and improved class-wise consistency—particularly in generating walkable roads, crosswalks, and other urban elements demanded by assistive technologies. The method was computationally efficient as well as consistent for domain adaptation under low-data settings.

5.4 Summary

This chapter presented a systematic account of the experimental workflow used to evaluate generative models for augmenting the SENSATION dataset. The Pix2PixHD model was first fine-tuned using pretrained weights from Cityscapes, followed by training from scratch to better adapt to the domain-specific characteristics of pedestrian navigation scenes. Detailed investigations into label remapping, discriminator configurations, and generator freezing strategies were conducted to enhance realism and class consistency. Subsequently, the Stable Diffusion + ControlNet pipeline was explored, both in zero-shot and fine-tuned settings. LoRA-based fine-tuning using colored masks yielded improved semantic fidelity and visual quality. Key hyperparameters such as guidance scale and controlnet conditioning strength were tuned to balance structural coherence and photorealism. The next chapter would highlight the comparison of results of experiments mentioned in this chapter.

Chapter 6

Evaluation of Results

Manual inspection of images produced as a result of *Experiment 1* showed poor-quality images as the pretrained model clearly struggled to adapt to the features of the new dataset. This was evident in the worsening quality of generated images after the global generator's layers were unfrozen. Hence, it only made sense to consider the test results of the Pix2PixHD model trained on augmented SENSATION dataset which showed realistic images being synthesized during its training process as shown in Fig.5.3. These results from the cGAN model were compared with those of the fine-tuned ControlNet model from *Experiment 3*. Noticeable good quality results while preserving the semantic structure of the mask were produced by the ControlNet model as seen is Fig.6.1. Despite the fact that the cGAN model could preserve the semantic structure of the masks used, it did not introduce diversify and multiply the dataset on a scale on which the ControlNet model did.

6.1 Quantitative Evaluation

To objectively assess the performance of the generative models used in this thesis, a series of quantitative metrics were employed. The evaluation focuses on comparing the image quality, diversity, and semantic consistency of the outputs produced by the best-performing Pix2PixHD model (trained from scratch) and the fine-tuned Stable Diffusion + ControlNet model.

6.1.1 Fréchet Inception Distance (FID)

The final dataset produced out of the ControlNet model had 15,596 images. Of these, 2000 random images were chosen to calculate this subset's FID. Similarly, 2000 images

Figure 6.1: Semantic Mask(left) and its Synthesized Image(right)

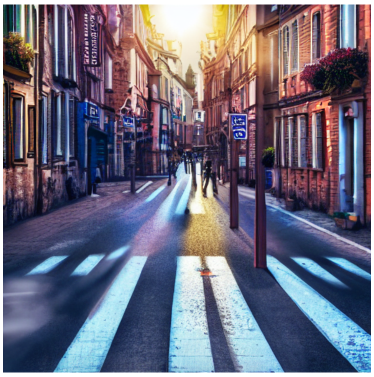
produced from the Pix2PixHD model were taken into account to calculate their FID. Despite lacking preservation of semantic structure, Stable Diffusion's outputs were high quality and worth comparing. 2000 images of the same were taken for the calculation of their FID. Table6.1 shows the mean FID scores of each model. It is immediately observable that Stable Diffusion shows the lowest FID but cannot be used in this case because it cannot preserve the semantic structure of the labels of the SENSATION dataset.

Table 6.1: Estimated FID Scores for Different Models

Model	**Estimated FID Score**
Pix2PixHD (Trained from Scratch)	63
Stable Diffusion (Pretrained)	17
Stable Diffusion + ControlNet (Fine-tuned with LoRA)	28

6.1.2 Redundancy Ratio

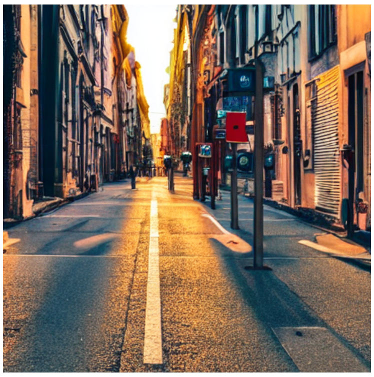
To assess the diversity of the generated dataset from the ControlNet model, the redundancy ratio of the images was computed. Among the 15,596 generated images, no duplicates or near-duplicates were detected, resulting in a redundancy ratio of **0.00%**. This indicates that the model produced a highly diverse set of outputs, which is particularly valuable for data augmentation purposes. The absence of redundancy confirms the dataset's potential to contribute meaningful variance to training pipelines. This can be visually inspected in Fig.6.2.



(a) Image 1

(b) Image 2

(c) Image 3



(d) Image 4

(e) Image 5

Figure 6.2: Comparison of 5 generated outputs from the model with the same semantic label but different text prompts

6.1.3 Segmentation-Based Evaluation of Semantic Consistency

To further assess the semantic realism and consistency of the synthetic images generated from the ControlNet model, a segmentation-based test was conducted. The common thought was to verify how accurately a pre-trained semantic segmenting model interprets the generated images with respect to the corresponding ground truth masks.

The ResNet101 backbone DeepLabV3 model pretrained on the COCO and VOC datasets was used as the segmentation backbone. Each synthetic image was passed through this segmentation model to produce a predicted semantic mask. These predicted masks were then compared to the original masks used for generation, which are used as the ground truth here.

To quantify the alignment between predicted and ground truth masks, mean Intersection over Union (mIoU) score was calculated across the dataset of 15k images.table6.2 shows the value obtained.

This enables cross-modal comparison across generation and segmentation pipelines and indirectly quantifies how realistic and correct the models mirror the underlying layout of classes. A better segmentation model could aid in better semantic mask outputs.

Table 6.2: Mean Intersection over Union (mIoU) Evaluation

Model	mIoU Score	Interpretation
Stable Diffusion + ControlNet	0.65	Good consistency; minor alignment issues

6.2 Qualitative Evaluation

6.2.1 Human Perceptual Study

With limited time and resources, 10 people were involved in a study to rate 25 images' preferences in terms of image quality between the images produced from the Pix2PixHD model trained entirely on the augmented SENSATION dataset and the images created by fine-tuning the Stable Diffusion and ControlNet model. The images were selected randomly. The 25 images presented were the same for each person and it was an unlimited time period survey implying that the participants were not restrained by time.Table6.3 summarises the results.

Table 6.3: Human Perceptual Preference Between Pix2PixHD and Stable Diffusion + ControlNet

Total Image Comparisons	Pix2PixHD (%)	Stable Diffusion + ControlNet (%)
250	10	90

6.3 Summary

The Results and Evaluation chapter assessed the performance of generative models used in this thesis, namely Pix2PixHD, Stable Diffusion, and Stable Diffusion with ControlNet, using both quantitative and qualitative metrics. Quantitative measurement was done using the FID and mean Intersection over Union (mIoU), providing a measure of the realism and semantic alignment of the generated images. Stable Diffusion models, particularly with ControlNet fine-tuning, significantly outperformed Pix2PixHD on photorealism and FID. In addition to this, a human perceptual test identified strong preference for the images from the diffusion-based models. Qualitative analysis and segmentation-based results also established the semantic integrity and diversity of the samples, demonstrating the effectiveness of diffusion-based approaches in generating structured synthetic data.

Chapter 7

Conclusion

7.1 Summary

This thesis investigated the role of generative models—specifically cGANs and diffusion models for the purpose of synthetic labeled data generation. The central motivation was to address limitations of existing datasets in assistive navigation for visually impaired individuals, particularly the lack of semantically rich images where sidewalks are centrally focused.

Since the main aim was dataset augmentation, efforts were directed towards maximising the augmentation process at every step from the application of pre-processing for augmentation to using the blurry output obtained from fine-tuning cGAN as input for the ControlNet model. A custom dataset, SENSATION, comprising 3,233 image-mask pairs was preprocessed through class-wise visualization and augmentation strategies. Two-stage augmentation, involving cropping underrepresented classes followed by geometric transformations, successfully balanced the dataset and improved representation diversity.

While modeling, Pix2PixHD was tried both by transfer learning from Cityscapes and full training from the augmenteded dataset. Various architectural tuning methods such as label remapping, discriminator scaling, and freezing generator layers were tried. While the results improved over time, the outputs lacked sufficient variety of texture and did not capture fine semantic details.

On the other hand, Stable Diffusion combined with ControlNet and fine-tuned using the LoRA method showed strong results in both semantic coherence and visual realism. Conditioning via segmentation masks, combined with strength and guidance parameters, allowed for high-level control over image outputs. Evaluation using FID scores and redundancy ratio (0.00%) confirmed that diffusion models produced a richer, more diverse synthetic dataset, outperforming cGAN-based approaches.

Towards the end, the initial goal of comparison of cGAN and diffusion techniques turned into comparison and integration of the two to achieve better results than their individual uses through the usage of blurry output from fine-tuning the cGAN as input for the ControlNet model.

7.2 Recommendations

Based on the experiments and results observed throughout this work, the following recommendations can be made:

- For use cases requiring **high photorealism and diversity**, diffusion-based models like Stable Diffusion with ControlNet are preferable to cGANs, especially when conditioning on segmentation masks.
- **Transfer learning with pretrained cGANs** requires careful semantic label alignment. Remapping label IDs to match the pretrained model's domain is crucial and despite the re-alignment the model struggles to adapt to new features.
- **Data visualization** before augmentation is vital for identifying class imbalance, guiding the creation of a robust and balanced training set.
- **Fine-tuning with LoRA** offers an efficient alternative for domain adaptation in diffusion models without retraining from scratch.
- **Hyperparameter tuning**, particularly discriminator count and global generator freezing in cGANs and control strength in diffusion models, significantly affects output quality and should be considered a core part of the experimentation.

7.3 Future Work

Several directions remain open for future exploration. The most important observation and suggested direction for work could be further integration of cGANs and ControlNet. This could be enhanced by studying the feature overlap between the dataset on which pretrained model was trained and the dataset used for downstreaming tasks.

The augmented dataset's image quality could be enhanced through various methods making it more suitable for training the navigation assistance system. Finally, expanding the diversity of generated prompts in text-to-image models could enhance realism even further. Applying similar synthesis strategies across multimodal datasets could advance the creation of complex virtual environments for simulation and assistive technology testing.

List of Abbreviations

SVM Support Vector Machine

NN Neural Network

ML Machine Learning

DS Dataset

GAN Generative Adversarial Network

cGAN Conditional Generative Adversarial Network

VAE Variational Autoencoder

DNN Deep Neural Network

AI Artificial Intelligence

FID Fréchet Inception Distance

VCI Visually Challenged Individual

ETA Electronic Travel Aid

CLIP Contrastive Language–Image Pretraining

GPU Graphics Processing Unit

TPU Tensor Processing Unit

SENSATION Sidewalk Environment Detection System for Assistive Navigation

DL Deep Learning

LDM Latent Diffusion Model

FAU Fredrich Alexander University

SDM Semantic Diffusion Model

LoRA Low-Rank Adaptation

List of Figures

List of Tables

Bibliography

[1] M. Cordts, M. Omran, S. Ramos, T. Rehfeld, M. Enzweiler, R. Benenson, U. Franke, S. Roth, and B. Schiele, *The Cityscapes Dataset for Semantic Urban Scene Understanding*, arXiv:1604.01685 [cs], Apr. 2016. DOI: 10.48550/arXiv.1604.01685. [Online]. Available: http://arxiv.org/abs/1604.01685 (visited on 05/02/2025).

[2] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks", in *Advances in Neural Information Processing Systems*, vol. 25, Curran Associates, Inc., 2012. [Online]. Available: https://papers.nips.cc/paper_files/paper/2012/hash/c399862d3b9d6b76c8436e924a68c45b-Abstract.html (visited on 04/30/2025).

[3] A. Antoniou, A. Storkey, and H. Edwards, *Data Augmentation Generative Adversarial Networks*, arXiv:1711.04340 [stat], Mar. 2018. DOI: 10.48550/arXiv.1711.04340. [Online]. Available: http://arxiv.org/abs/1711.04340 (visited on 05/01/2025).

[4] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, *Image-to-Image Translation with Conditional Adversarial Networks*, arXiv:1611.07004 [cs], Nov. 2018. [Online]. Available: http://arxiv.org/abs/1611.07004 (visited on 07/07/2024).

[5] T.-C. Wang, M.-Y. Liu, J.-Y. Zhu, A. Tao, J. Kautz, and B. Catanzaro, *High-Resolution Image Synthesis and Semantic Manipulation with Conditional GANs*, arXiv:1711.11585, Aug. 2018. [Online]. Available: http://arxiv.org/abs/1711.11585 (visited on 10/27/2024).

[6] T. Park, M.-Y. Liu, T.-C. Wang, and J.-Y. Zhu, *Semantic Image Synthesis with Spatially-Adaptive Normalization*, arXiv:1903.07291, Nov. 2019. [Online]. Available: http://arxiv.org/abs/1903.07291 (visited on 10/15/2024).

[7] L. Zhang, A. Rao, and M. Agrawala, *Adding Conditional Control to Text-to-Image Diffusion Models*, arXiv:2302.05543 [cs], Nov. 2023. DOI: 10.48550/arXiv.2302.05543. [Online]. Available: http://arxiv.org/abs/2302.05543 (visited on 04/18/2025).

[8] W. Wang, J. Bao, W. Zhou, D. Chen, D. Chen, L. Yuan, and H. Li, *Semantic Image Synthesis via Diffusion Models*, arXiv:2207.00050, Nov. 2022. [Online]. Available: http://arxiv.org/abs/2207.00050 (visited on 11/14/2024).

[9] I. Goodfellow, Y. Bengio, and A. Courville, *Deep learning* (Adaptive computation and machine learning). Cambridge, Massachusetts: The MIT Press, 2016, ISBN: 978-0-262-03561-3.

[10] J. Schmidhuber, "Deep learning in neural networks: An overview", en, *Neural Networks*, vol. 61, pp. 85–117, Jan. 2015, Publisher: Elsevier BV, ISSN: 0893-6080. DOI: 10.1016/j.neunet.2014.09.003. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0893608014002135 (visited on 04/30/2025).

[11] L. Alzubaidi, J. Bai, A. Al-Sabaawi, J. Santamaría, A. S. Albahri, B. S. N. Al-dabbagh, M. A. Fadhel, M. Manoufali, J. Zhang, A. H. Al-Timemy, Y. Duan, A. Abdullah, L. Farhan, Y. Lu, A. Gupta, F. Albu, A. Abbosh, and Y. Gu, "A survey on deep learning tools dealing with data scarcity: Definitions, challenges, solutions, tips, and applications", en, *Journal of Big Data*, vol. 10, no. 1, Apr. 2023, Publisher: Springer Science and Business Media LLC, ISSN: 2196-1115. DOI: 10.1186/s40537-023-00727-2. [Online]. Available: https://journalofbigdata.springeropen.com/articles/10.1186/s40537-023-00727-2 (visited on 04/30/2025).

[12] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning", en, *Nature*, vol. 521, no. 7553, pp. 436–444, May 2015, Publisher: Springer Science and Business Media LLC, ISSN: 0028-0836, 1476-4687. DOI: 10.1038/nature14539. [Online]. Available: https://www.nature.com/articles/nature14539 (visited on 04/30/2025).

[13] Bishop and M. Christopher, *Pattern Recognition and Machine Learning*. Springer New York, 2006, ISBN: 978-0-387-31073-2. DOI: 10.1007/978-0-387-45528-0. [Online]. Available: https://link.springer.com/deleted (visited on 05/01/2025).

[14] S. Yang, W. Xiao, M. Zhang, S. Guo, J. Zhao, and F. Shen, *Image Data Augmentation for Deep Learning: A Survey*, arXiv:2204.08610 [cs], Nov. 2023. DOI: 10.48550/arXiv.2204.08610. [Online]. Available: http://arxiv.org/abs/2204.08610 (visited on 05/01/2025).

[15] J. Choi, C. Lee, D. Lee, and H. Jung, "SalfMix: A Novel Single Image-Based Data Augmentation Technique Using a Saliency Map", en, *Sensors*, vol. 21, no. 24, p. 8444, Dec. 2021, Publisher: MDPI AG, ISSN: 1424-8220. DOI: 10.3390/s21248444. [Online]. Available: https://www.mdpi.com/1424-8220/21/24/8444 (visited on 05/01/2025).

[16] J. Wang, S. Wang, and Y. Zhang, "Artificial intelligence for visually impaired", *Displays*, vol. 77, p. 102 391, 2023, ISSN: 0141-9382. DOI: https://doi.org/10.1016/j.displa.2023.102391. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0141938223000240.

[17] P. Naayini, P. K. Myakala, C. Bura, A. K. Jonnalagadda, and S. Kamatala, *AI-Powered Assistive Technologies for Visual Impairment*, arXiv:2503.15494 [cs], Jan. 2025. DOI: 10.48550/arXiv.2503.15494. [Online]. Available: http://arxiv.org/abs/2503.15494 (visited on 05/01/2025).

[18] X. Zhang, X. Yao, Y. Zhu, and F. Hu, "An ARCore Based User Centric Assistive Navigation System for Visually Impaired People", en, *Applied Sciences*, vol. 9, no. 5, p. 989, Mar. 2019, Publisher: MDPI AG, ISSN: 2076-3417. DOI: 10.3390/app9050989. [Online]. Available: https://www.mdpi.com/2076-3417/9/5/989 (visited on 05/01/2025).

[19] I. Goodfellow, *NIPS 2016 Tutorial: Generative Adversarial Networks*, arXiv:1701.00160 [cs], Apr. 2017. DOI: 10.48550/arXiv.1701.00160. [Online]. Available: http://arxiv.org/abs/1701.00160 (visited on 04/30/2025).

[20] X. Guo and Y. Chen, *Generative AI for Synthetic Data Generation: Methods, Challenges and the Future*, arXiv:2403.04190 [cs], Mar. 2024. DOI: 10.48550/arXiv.2403.04190. [Online]. Available: http://arxiv.org/abs/2403.04190 (visited on 05/01/2025).

[21] K. P. Murphy, *Machine learning: a probabilistic perspective* (Adaptive computation and machine learning series). Cambridge, MA: MIT Press, 2012, ISBN: 978-0-262-01802-9.

[22] I. J. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, *Generative Adversarial Networks*, arXiv:1406.2661 [stat], Jun. 2014. DOI: 10.48550/arXiv.1406.2661. [Online]. Available: http://arxiv.org/abs/1406.2661 (visited on 05/01/2025).

[23] C. Little, M. Elliot, R. Allmendinger, and S. S. Samani, *Generative Adversarial Networks for Synthetic Data Generation: A Comparative Study*, Version Number: 1, 2021. DOI: 10.48550/ARXIV.2112.01925. [Online]. Available: https://arxiv.org/abs/2112.01925 (visited on 05/01/2025).

[24] M. Arjovsky, S. Chintala, and L. Bottou, *Wasserstein GAN*, arXiv:1701.07875 [stat], Dec. 2017. DOI: 10.48550/arXiv.1701.07875. [Online]. Available: http://arxiv.org/abs/1701.07875 (visited on 05/01/2025).

[25] C. Tan, F. Sun, T. Kong, W. Zhang, C. Yang, and C. Liu, *A Survey on Deep Transfer Learning*, arXiv:1808.01974 [cs], Aug. 2018. DOI: 10.48550/arXiv.1808.01974. [Online]. Available: http://arxiv.org/abs/1808.01974 (visited on 05/01/2025).

[26] J. Sohl-Dickstein, E. A. Weiss, N. Maheswaranathan, and S. Ganguli, *Deep Unsupervised Learning using Nonequilibrium Thermodynamics*, arXiv:1503.03585 [cs], Nov. 2015. DOI: 10.48550/arXiv.1503.03585. [Online]. Available: http://arxiv.org/abs/1503.03585 (visited on 05/01/2025).

[27] J. Ho, A. Jain, and P. Abbeel, *Denoising Diffusion Probabilistic Models*, arXiv:2006.11239, Dec. 2020. [Online]. Available: http://arxiv.org/abs/2006.11239 (visited on 11/17/2024).

[28] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, *GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium*, arXiv:1706.08500 [cs], Jan. 2018. DOI: 10.48550/arXiv.1706.08500. [Online]. Available: http://arxiv.org/abs/1706.08500 (visited on 05/01/2025).

[29] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, *LoRA: Low-Rank Adaptation of Large Language Models*, arXiv:2106.09685 [cs], Oct. 2021. DOI: 10.48550/arXiv.2106.09685. [Online]. Available: http://arxiv.org/abs/2106.09685 (visited on 05/05/2025).

[30] M. Mirza and S. Osindero, *Conditional Generative Adversarial Nets*, arXiv:1411.1784, Nov. 2014. [Online]. Available: http://arxiv.org/abs/1411.1784 (visited on 10/15/2024).

[31] P. Dhariwal and A. Nichol, *Diffusion Models Beat GANs on Image Synthesis*, arXiv:2105.05233, Jun. 2021. [Online]. Available: http://arxiv.org/abs/2105.05233 (visited on 11/16/2024).

[32] A. Ulhaq and N. Akhtar, *Efficient Diffusion Models for Vision: A Survey*, arXiv:2210.09292 [cs], Mar. 2024. DOI: 10.48550/arXiv.2210.09292. [Online]. Available: http://arxiv.org/abs/2210.09292 (visited on 05/03/2025).

[33] A. Nichol and P. Dhariwal, *Improved Denoising Diffusion Probabilistic Models*, arXiv:2102.09672 [cs], Feb. 2021. DOI: 10.48550/arXiv.2102.09672. [Online]. Available: http://arxiv.org/abs/2102.09672 (visited on 05/04/2025).

[34] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, *High-Resolution Image Synthesis with Latent Diffusion Models*, arXiv:2112.10752 [cs], Apr. 2022. DOI: 10.48550/arXiv.2112.10752. [Online]. Available: http://arxiv.org/abs/2112.10752 (visited on 03/23/2025).

[35] C. Saharia, W. Chan, H. Chang, C. A. Lee, J. Ho, T. Salimans, D. J. Fleet, and M. Norouzi, *Palette: Image-to-Image Diffusion Models*, arXiv:2111.05826, May 2022. [Online]. Available: http://arxiv.org/abs/2111.05826 (visited on 11/03/2024).

[36] D. Cortizo, L. Fleta, and D. Tascon, *DarCoF/Idd-pix2pix*, original-date: 2021-04-29T18:28:52Z, May 2024. [Online]. Available: https://github.com/DarCoF/Idd-pix2pix (visited on 05/05/2025).

[37] N. Ruiz, Y. Li, V. Jampani, Y. Pritch, M. Rubinstein, and K. Aberman, *DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation*, arXiv:2208.12242 [cs], Mar. 2023. DOI: 10.48550/arXiv.2208.12242. [Online]. Available: http://arxiv.org/abs/2208.12242 (visited on 05/05/2025).

[38] R. Gal, Y. Alaluf, Y. Atzmon, O. Patashnik, A. H. Bermano, G. Chechik, and D. Cohen-Or, *An Image is Worth One Word: Personalizing Text-to-Image Generation using Textual Inversion*, arXiv:2208.01618 [cs], Aug. 2022. DOI: 10.48550/arXiv.2208.01618. [Online]. Available: http://arxiv.org/abs/2208.01618 (visited on 05/05/2025).

[39] C. Schuhmann, R. Beaumont, R. Vencu, C. Gordon, R. Wightman, M. Cherti, T. Coombes, A. Katta, C. Mullis, M. Wortsman, P. Schramowski, S. Kundurthy, K. Crowson, L. Schmidt, R. Kaczmarczyk, and J. Jitsev, *LAION-5B: An open large-scale dataset for training next generation image-text models*, arXiv:2210.08402 [cs], Oct. 2022. DOI: 10.48550/arXiv.2210.08402. [Online]. Available: http://arxiv.org/abs/2210.08402 (visited on 05/05/2025).

[40] T.-C. Wang, M.-Y. Liu, J.-Y. Zhu, A. Tao, J. Kautz, and B. Catanzaro, *NVIDIA/pix2pixHD: Synthesizing and manipulating 2048x1024 images with conditional GANs*. [Online]. Available: https://github.com/NVIDIA/pix2pixHD/tree/master (visited on 05/11/2025).

[41] W. Wang, J. Bao, W. Zhou, D. Chen, L. Yuan, and H. Li, *WeilunWang/semantic-diffusion-model: Official Implementation of Semantic Image Synthesis via Diffusion Models*. [Online]. Available: https://github.com/WeilunWang/semantic-diffusion-model?tab=readme-ov-file (visited on 05/12/2025).

[42] B. Zhou, H. Zhao, X. Puig, T. Xiao, S. Fidler, A. Barriuso, and A. Torralba, *Semantic Understanding of Scenes through the ADE20K Dataset*, arXiv:1608.05442 [cs], Oct. 2018. DOI: 10.48550/arXiv.1608.05442. [Online]. Available: http://arxiv.org/abs/1608.05442 (visited on 05/12/2025).

[43] *Hugging Face – The AI community building the future.* en-DE, May 2025. [Online]. Available: https://huggingface.co/ (visited on 05/12/2025).